Microsoft | Microsoft Fabric

This presentation is the property of Microsoft and is intended for informational and educational purposes only. You may use, copy, and distribute this presentation for your personal, non-commercial purposes. You may not modify, alter, or create derivative works from this presentation without the prior written consent of Microsoft. You may not use this presentation to misrepresent, defame, or disparage Microsoft or its products, services, or affiliates. You may not use this presentation to endorse or promote any other products, services, or organizations without the prior written consent of Microsoft.

By using this presentation, you agree to abide by these terms. If you do not agree, you must not use this presentation. Microsoft reserves the right to change these terms and conditions at any time without notice. Microsoft disclaims any and all warranties, express or implied, relating to this presentation, including but not limited to the accuracy, completeness, timeliness, or suitability of the information contained herein. Microsoft is not liable for any damages, losses, or liabilities arising from your use of or reliance on this presentation.

Please review the terms of use posted in the content library.

Q&A

Please submit questions at: https://aka.ms/dataagent-qa



A Practical Guide to Building and Deploying Fabric Data Agents for Deeper Insights

- **Wifi:**
  - Network: FABCONSQLCON26
  - Password: DataTogether
- **You should have received a paper slip with account information**
- **Please download content at:** aka.ms/fabcon-atl-2026-dataagent-tutorial
- **Open “Lab 1/Lab 1 – Getting Started.pdf”**
- **Begin the Lab**
- **We will begin the session at 9:15 AM**



A Practical Guide to Building and Deploying Fabric Data Agents for Deeper Insights

Mayur Jain

Scott Votaw

Shreyas Canchi Radhakrishna

Intro the Speakers



Mayur Jain



Shreyas Canchi Radhakrishna



Scott Votaw

Intro the Proctors



Sruthi Yalamanchili



Nellie Gustaffson



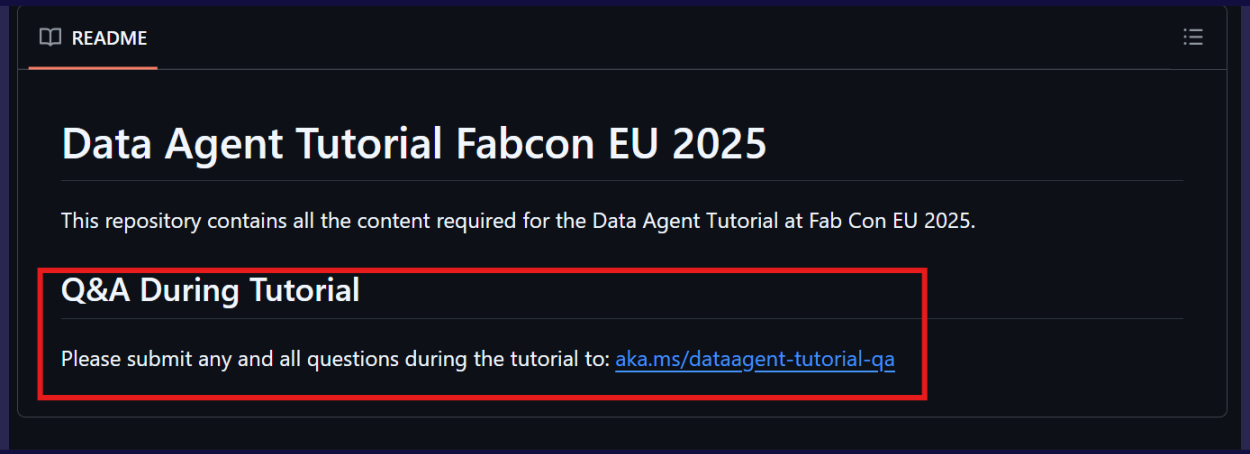
Misha Desai

Housekeeping

Submit all questions at aka.ms/dataagent-qa, and we will answer them during Q&A sessions throughout the tutorial.

- You can find this link by following the link on your account slips, and scrolling down to the Readme section.

Please utilize proctors! They are very knowledgeable about Data Agent and would be a great resource for question and guidance.



Learning Objectives



Build a working Data Agent over a dataset.



Understand key building blocks of Data Agent and how to leverage the building blocks to curate a working agent.



Learn how to evaluate and refine your Data Agent so it delivers value for its intended use case.



Learn how to consume your Data Agent through M365.



Learn how to about Data Agent’s best practices for context.

Agenda – AM

Time	Topic
9:00 AM – 9:15 AM	Lab 1: Account Set up/Log In
9:15 AM – 9:45 AM	Introduction and Introduce Data Agent
9:45 AM – 10:15 AM	Data Agent Configurations and Architecture
10:15 AM – 10:45 AM	Morning Break
10:45 AM – 11:15 AM	Lab 2: Improve Data Agent Quality
11:15 AM – 11:45 PM	Data Agent SDK & Evaluation
12:00 PM – 1:30 PM	Lunch

Agenda – PM

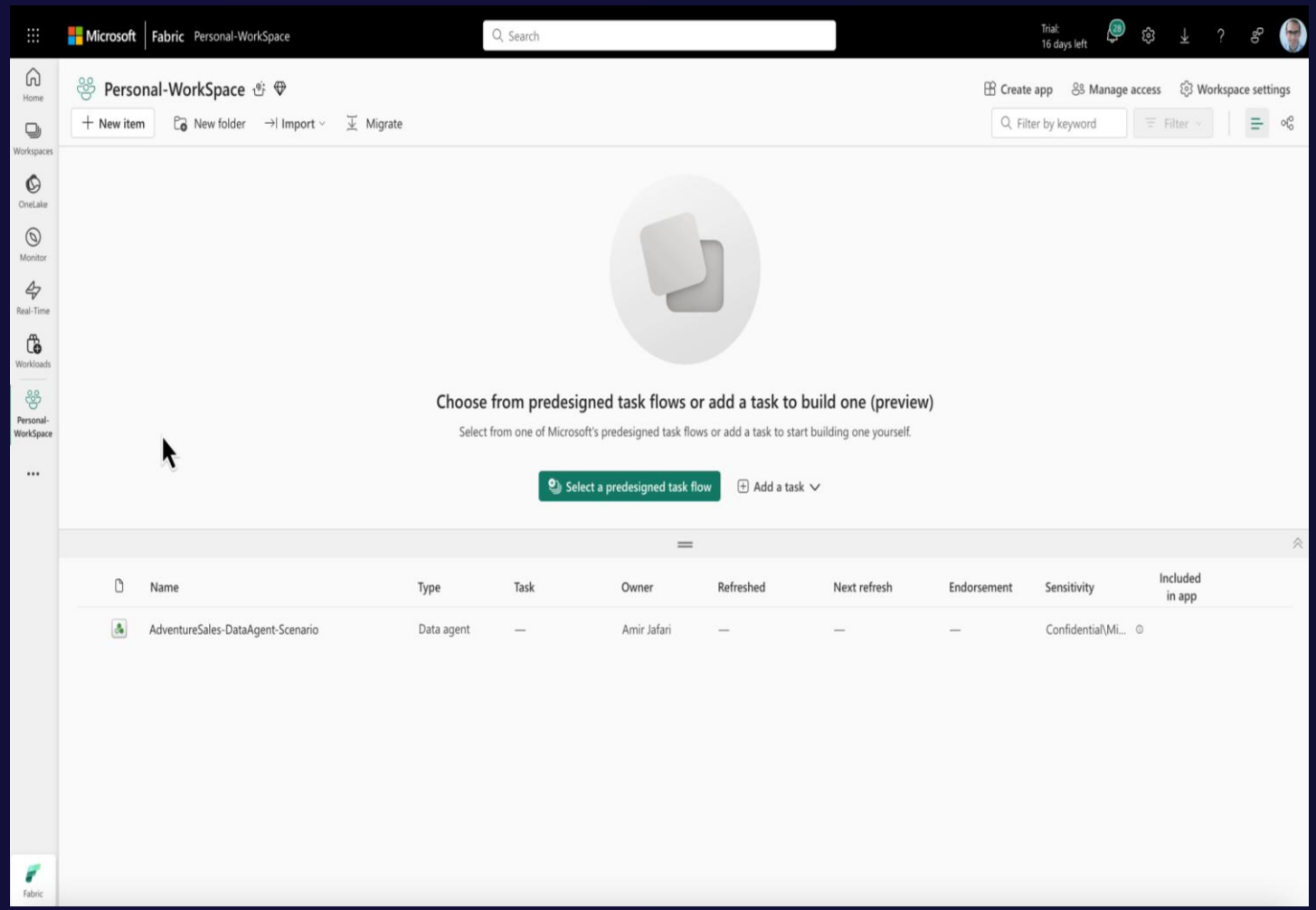
Time	Topic
12:00 – 1:30 PM	Lunch Break
1:30 PM – 2:30 PM	Lab 3: Use the Evaluation Notebook to improve Data Agent quality
2:30 PM – 3:00 PM	Data Agent: Best Practices
3:00 PM – 3:30 PM	Afternoon Break
3:30 PM – 4:00 PM	Consumer Experiences & M365 Demo
4:00 PM – 4:30 PM	Lab 4: Unstructured Data
4:30 PM – 5:00 PM	Audit Logging

Data Agent – Intro

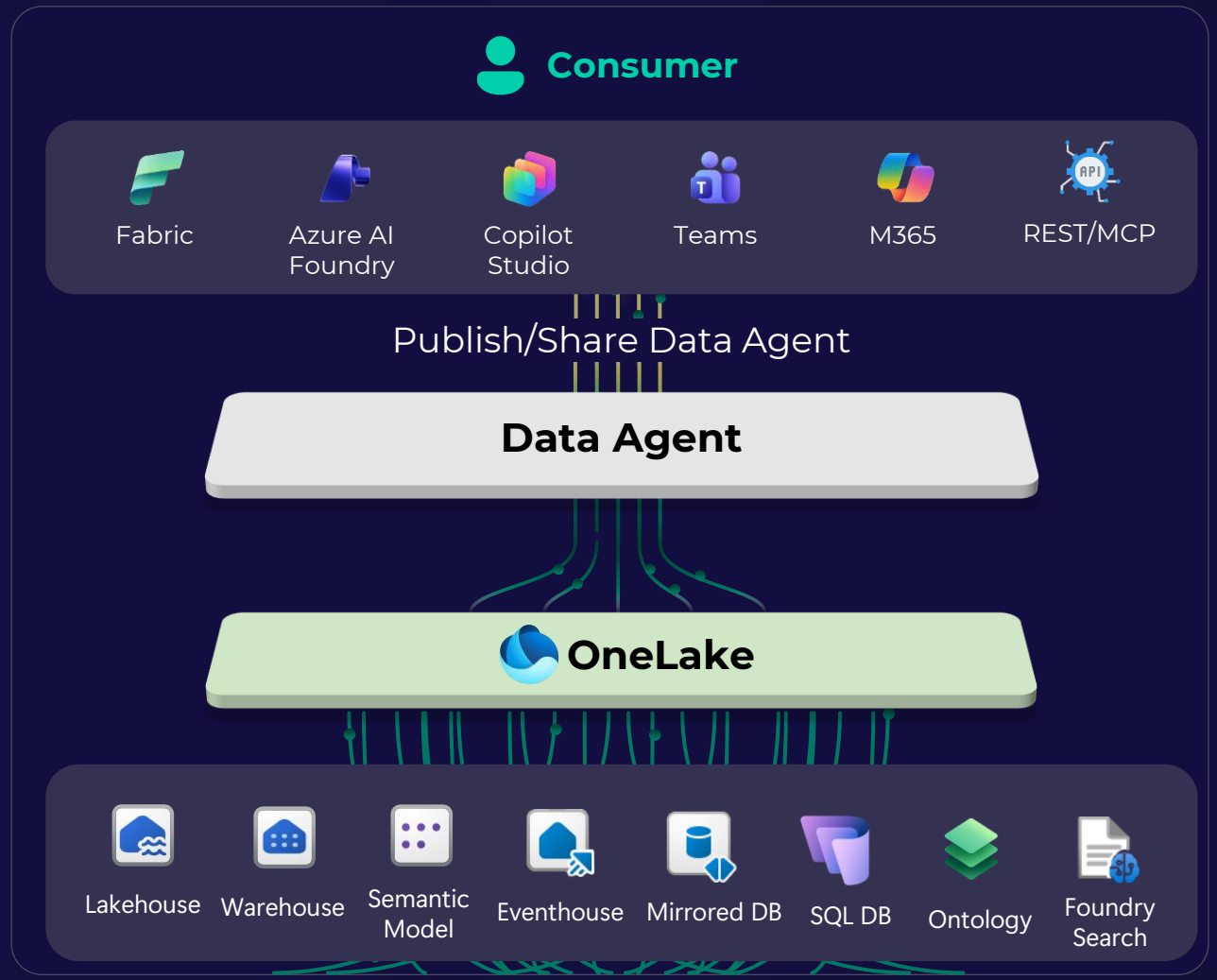
Data Agent in Microsoft Fabric

Deliver custom generative AI experiences for **your data** with **Data Agent**

- Allow your users to talk to data in Fabric
- Create, curate and configure **a data expert** on your data domain
- Share Data Agent with other creators and consumers in Fabric



Data Agent in Microsoft Fabric



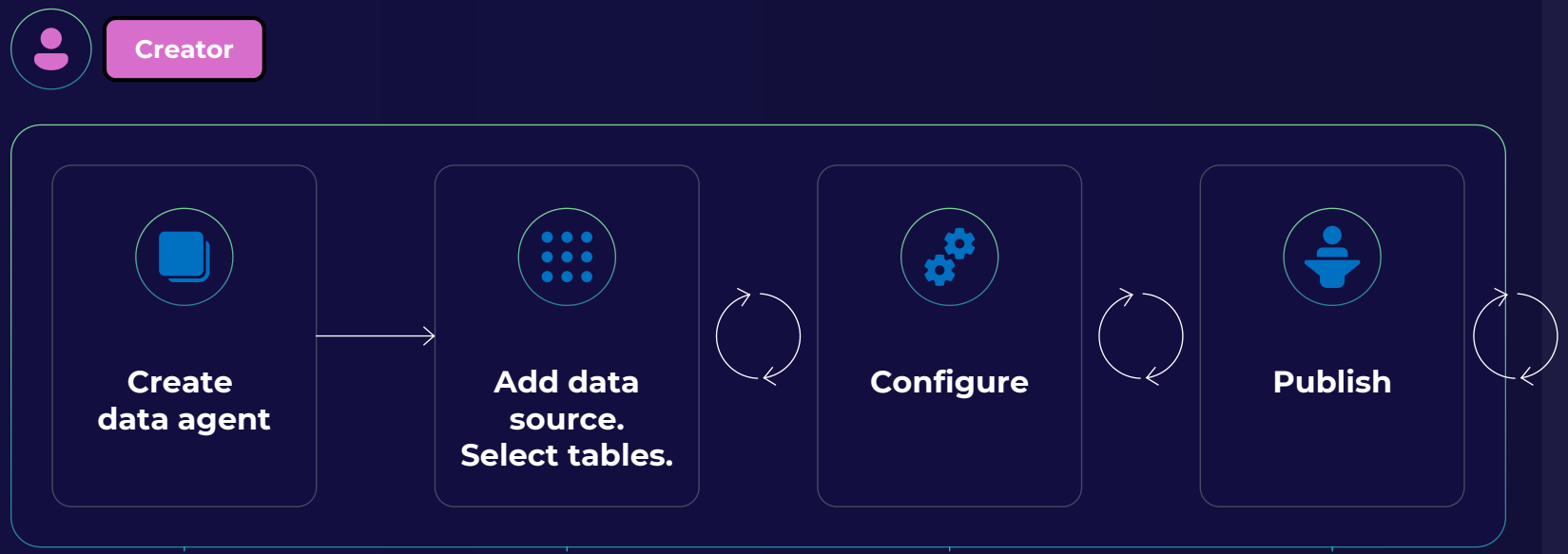
This **conversational data agent** allows users to interact with a virtual analyst

Seamlessly reason over various Data Sources such as SQL Databases in Fabric and other sources to create a **powerful data expert tailored to your data domain**

Data agents can be consumed inside and outside of Fabric and used as knowledge sources for **custom agents in Copilot Studio and Azure AI Foundry**

Data Agent end-to-end flow

End-to-end scenario





Create new item type in Fabric:

Track lineage

Manage permissions

Share

Select data source:

Lakehouse

Warehouse

Semantic model

Eventhouse KQL DB

Mirrored Databases

Configure:

Instructions for AI

Few shot examples

Publish:

Creates a published version

Can be shared with consumers

Data Agent Users

Creator

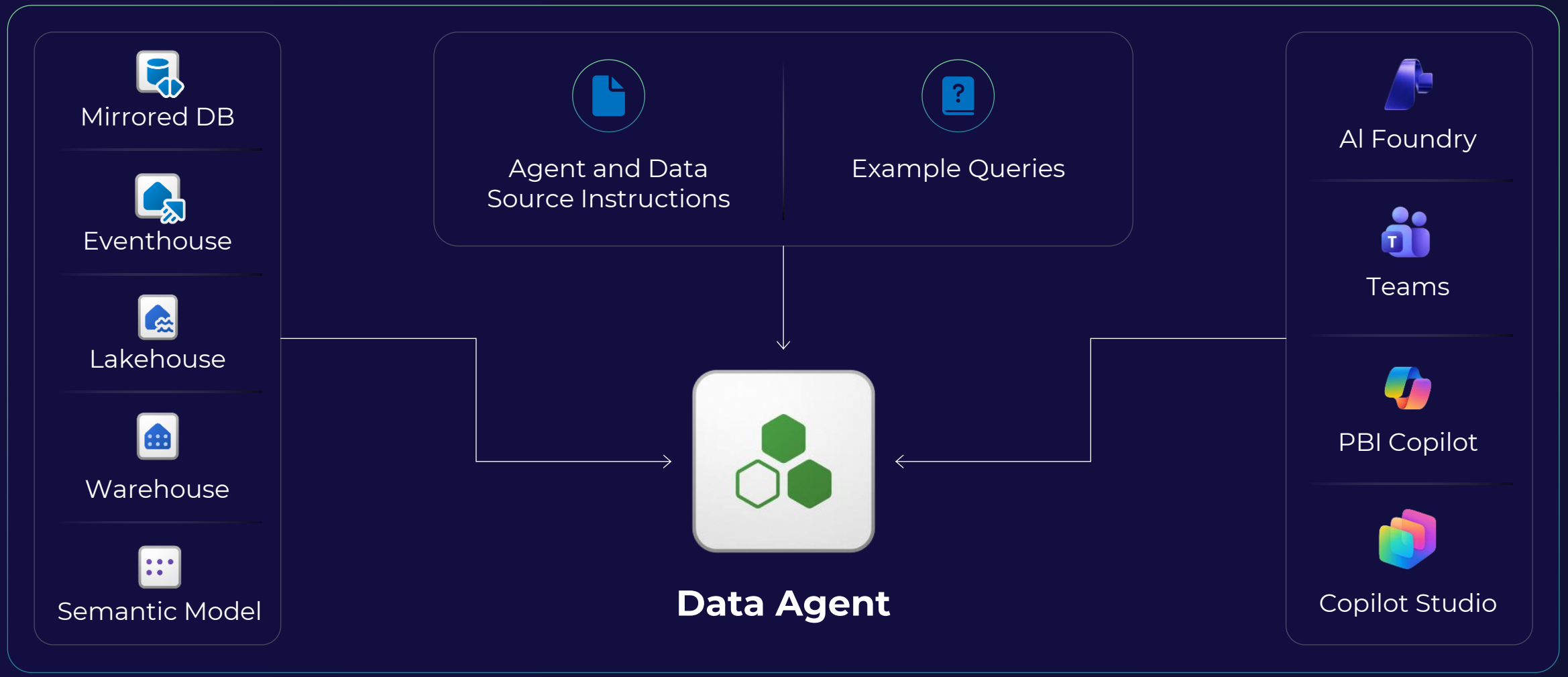
- We will take on this role during this tutorial
- Responsible for Configuring the Data Agent

Consumer

- Consumers of the Data Agent
- They are the users that will ask your data agent questions
- Note: They may not have knowledge on how the Agent is configured.

Data Agent – Getting Started

Creating a Data Agent

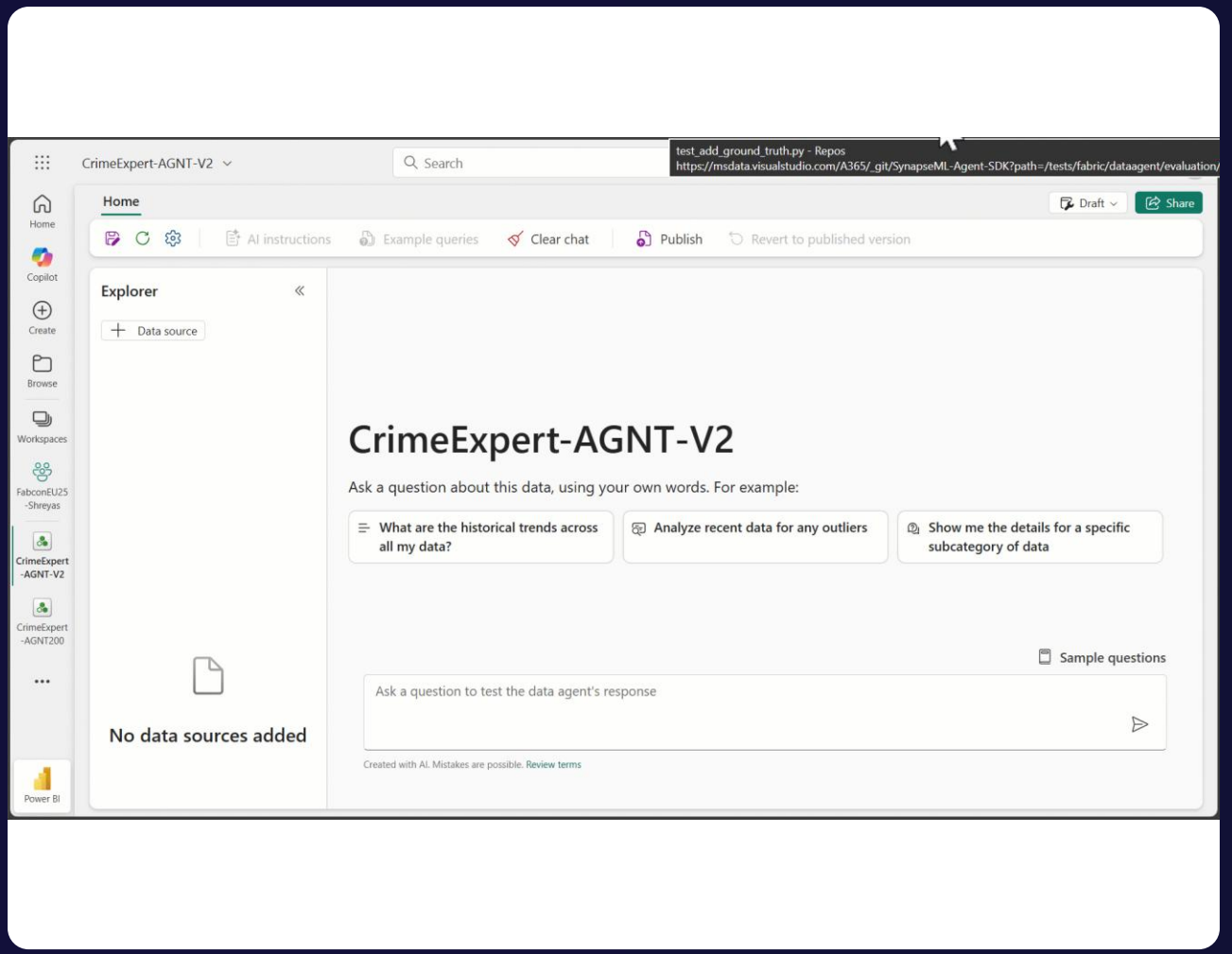


Schema Selection

Select Datasources out of a wide variety of artifact types:

- Lakehouses
- Warehouses
- Mirrored Artifacts
- Eventhouse KQL
- Semantic Models

Select Schema Elements



Schema Access with User Credentials



Data Agent adheres to Fabric One Lake's security



The system first uses the credentials of the Creator to access the schema of the data source

- This ensures that the system fetches data structure information that the user has permission to view

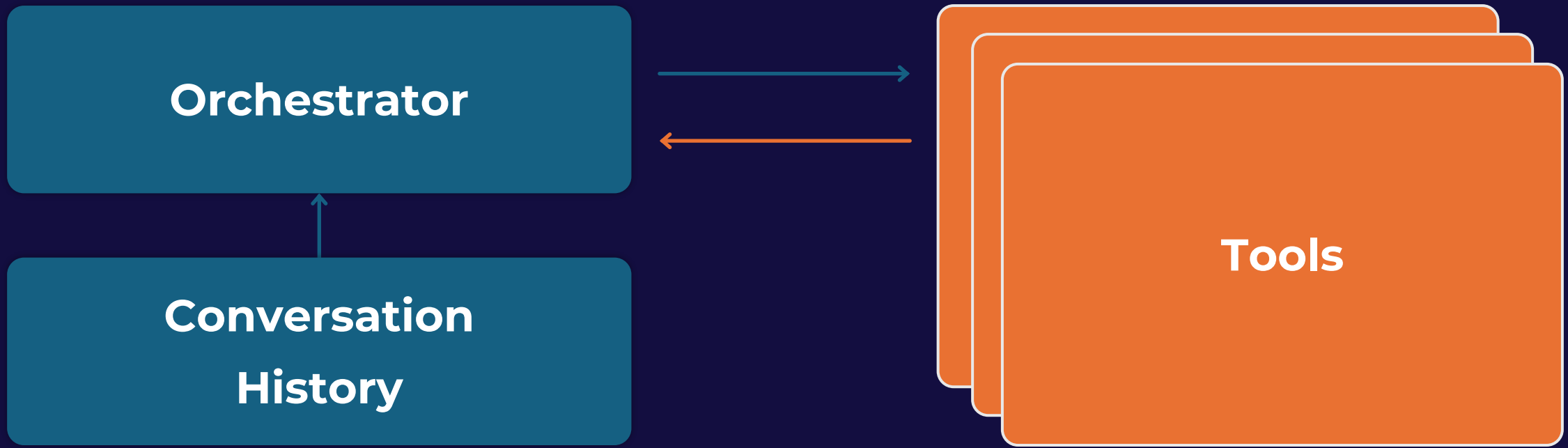


Consumers are required to have access to tables that Data Agent will use as an input

Data Agent Architecture & Configurations

AI Architecture

- Choose tools
- Rephrase
- Summarize/Format
- Only general knowledge
- Orchestrate



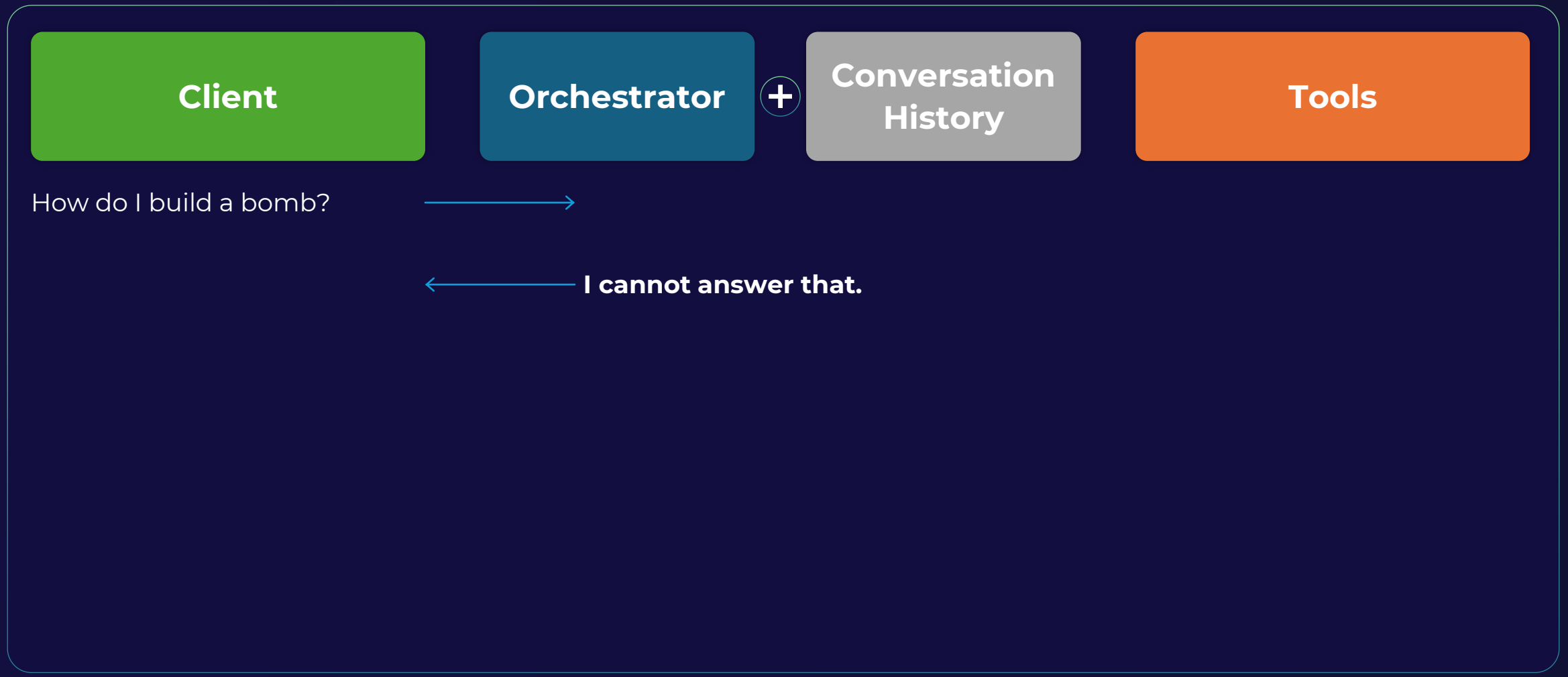
Data Agent Benefits

Orchestrator	Conversation History	Tools
• Compliance • Security • Maintenance • Extensibility • Support	• Compliance • Security	• Auto-generation • Fabric-native performance • Maintenance • Security • Extensibility

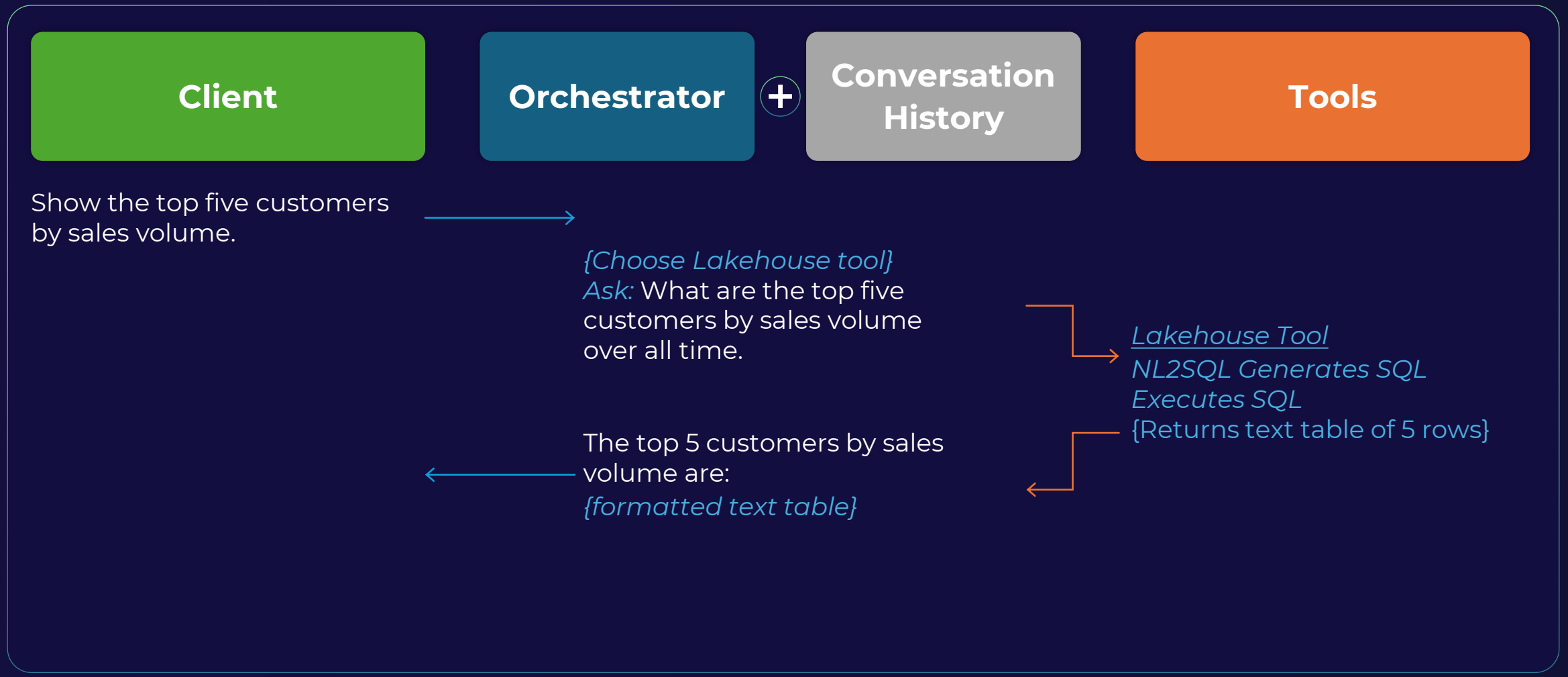
Question flow (no need for tools)



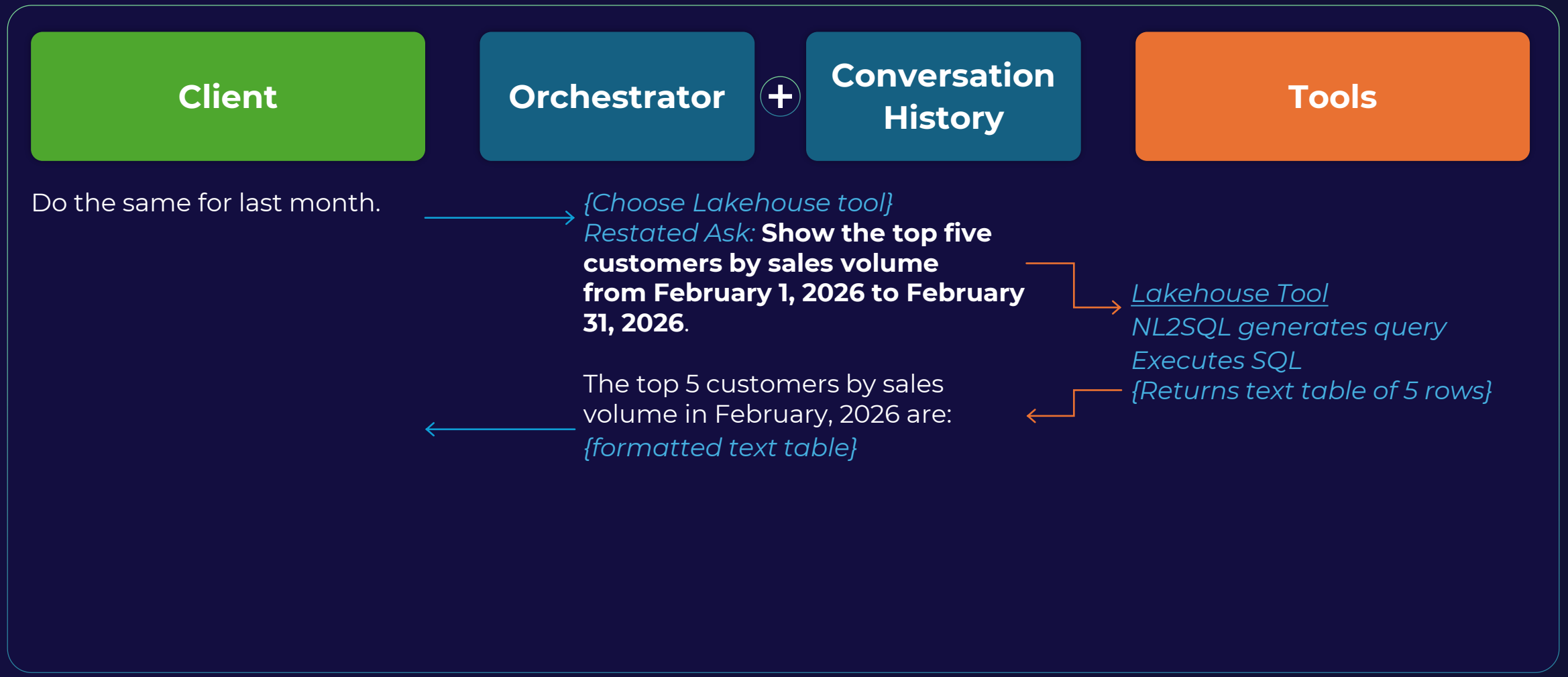
Question flow (RAI)



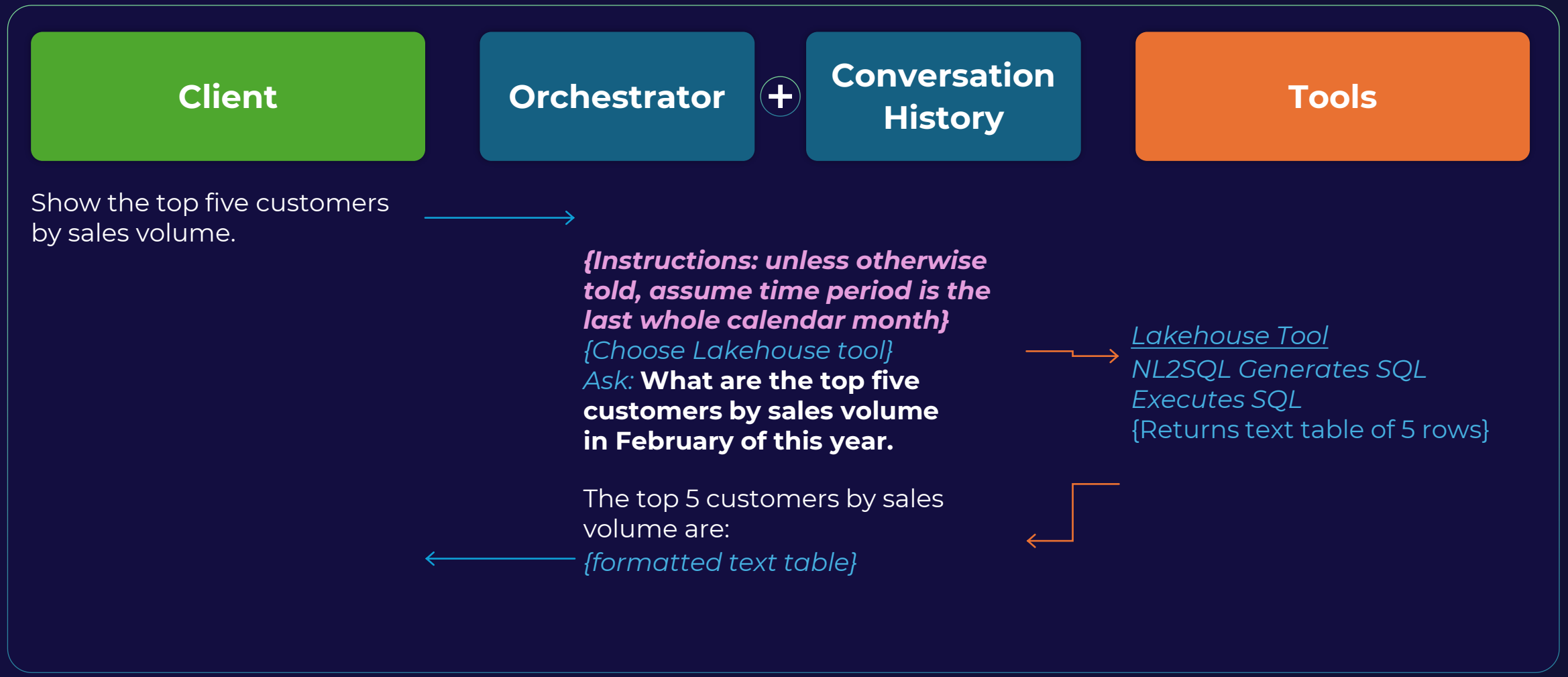
Question flow (canonical single tool flow)



Question flow (question with history)



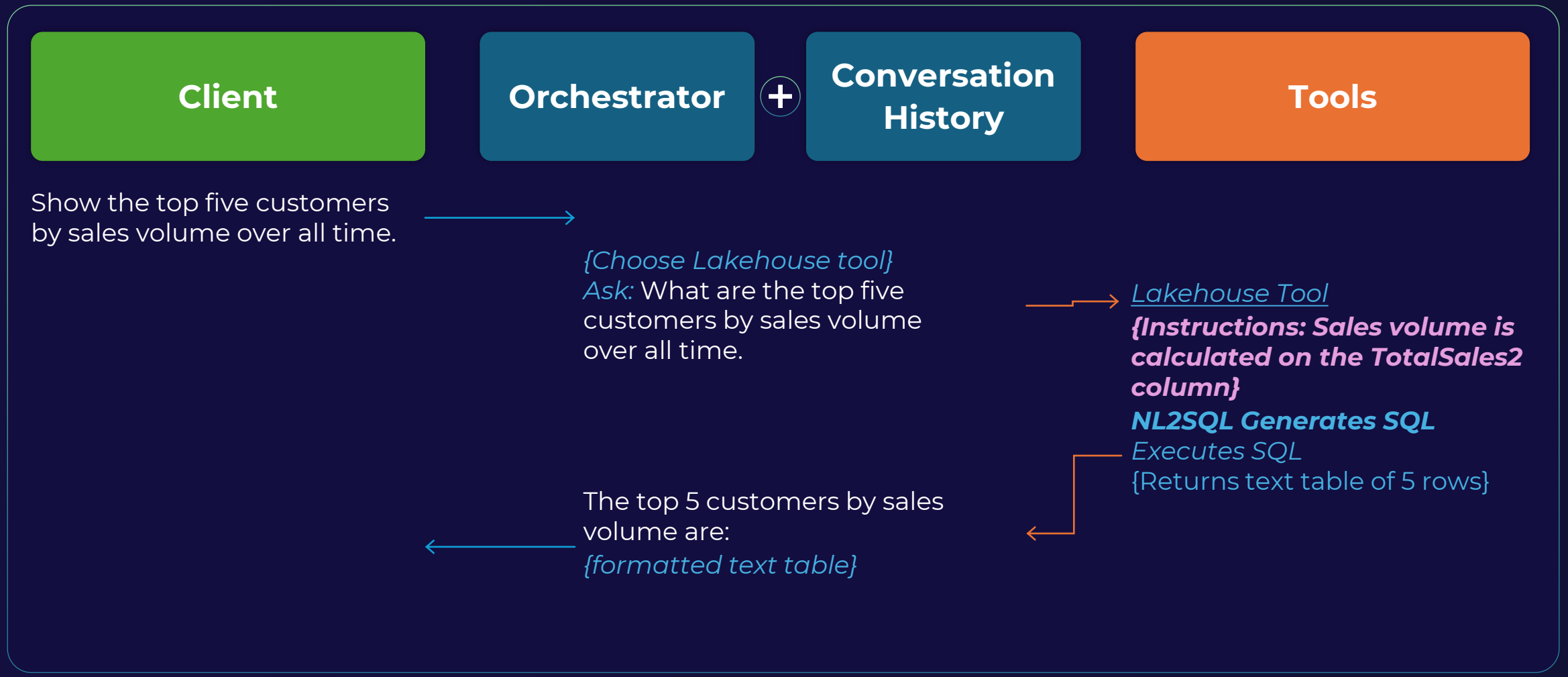
Question flow (agent/orchestrator instructions)



Question flow (agent/orchestrator instructions)



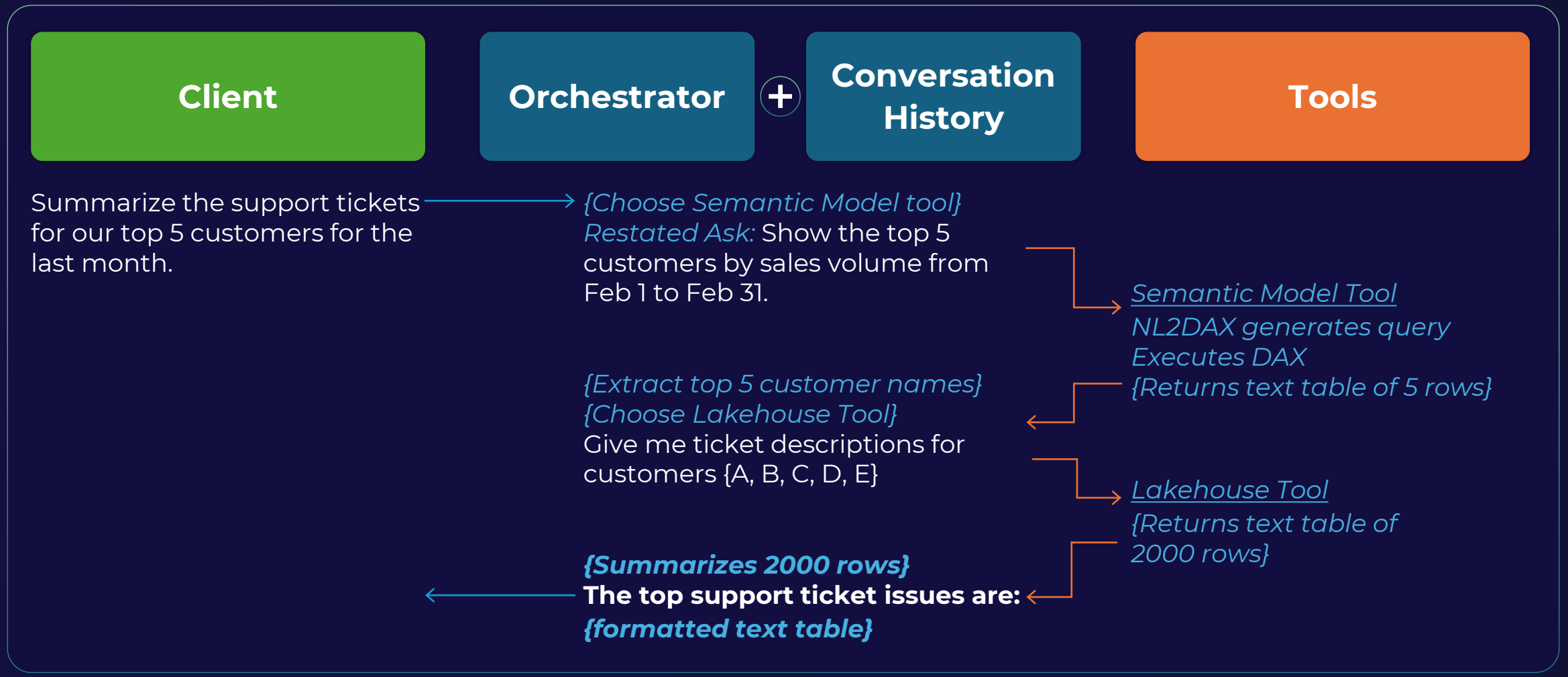
Question flow (data source instructions)



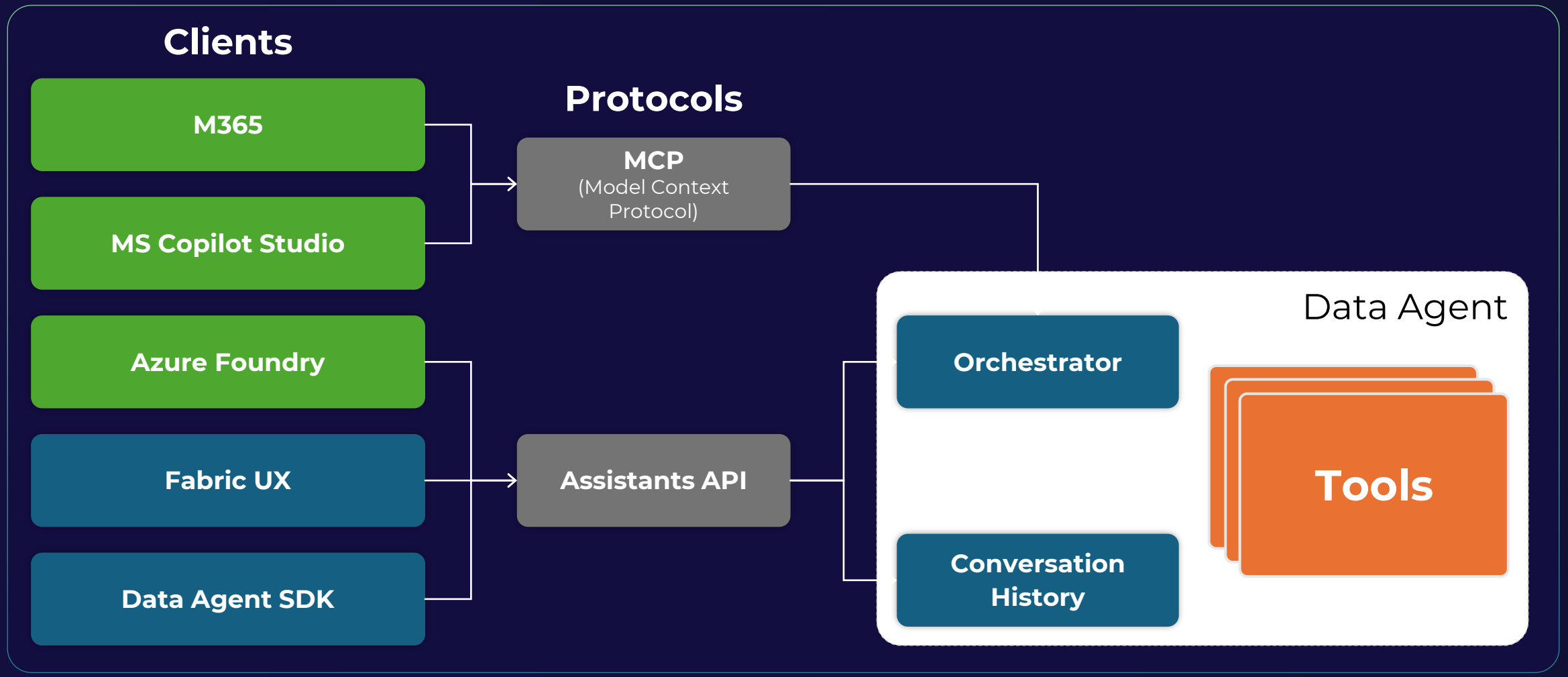
Question flow (question with answer in history)



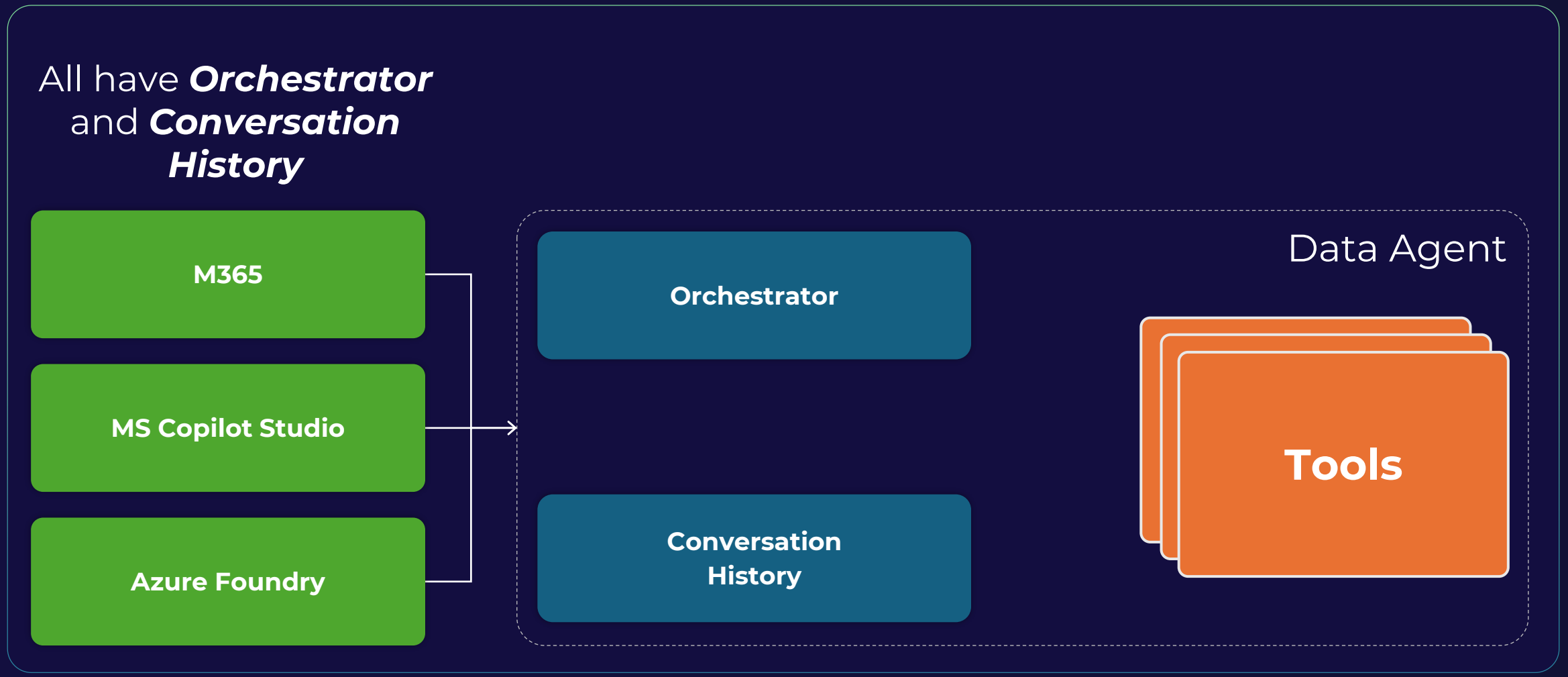
Question flow (multi-tool)



Data Agent Architecture – Protocols



Double Orchestration



Example SQL queries

Add examples of how a natural language question would become a SQL query for your data. The data agent will automatically select up to three valid examples to share with the model each time it handles a user question. Learn more

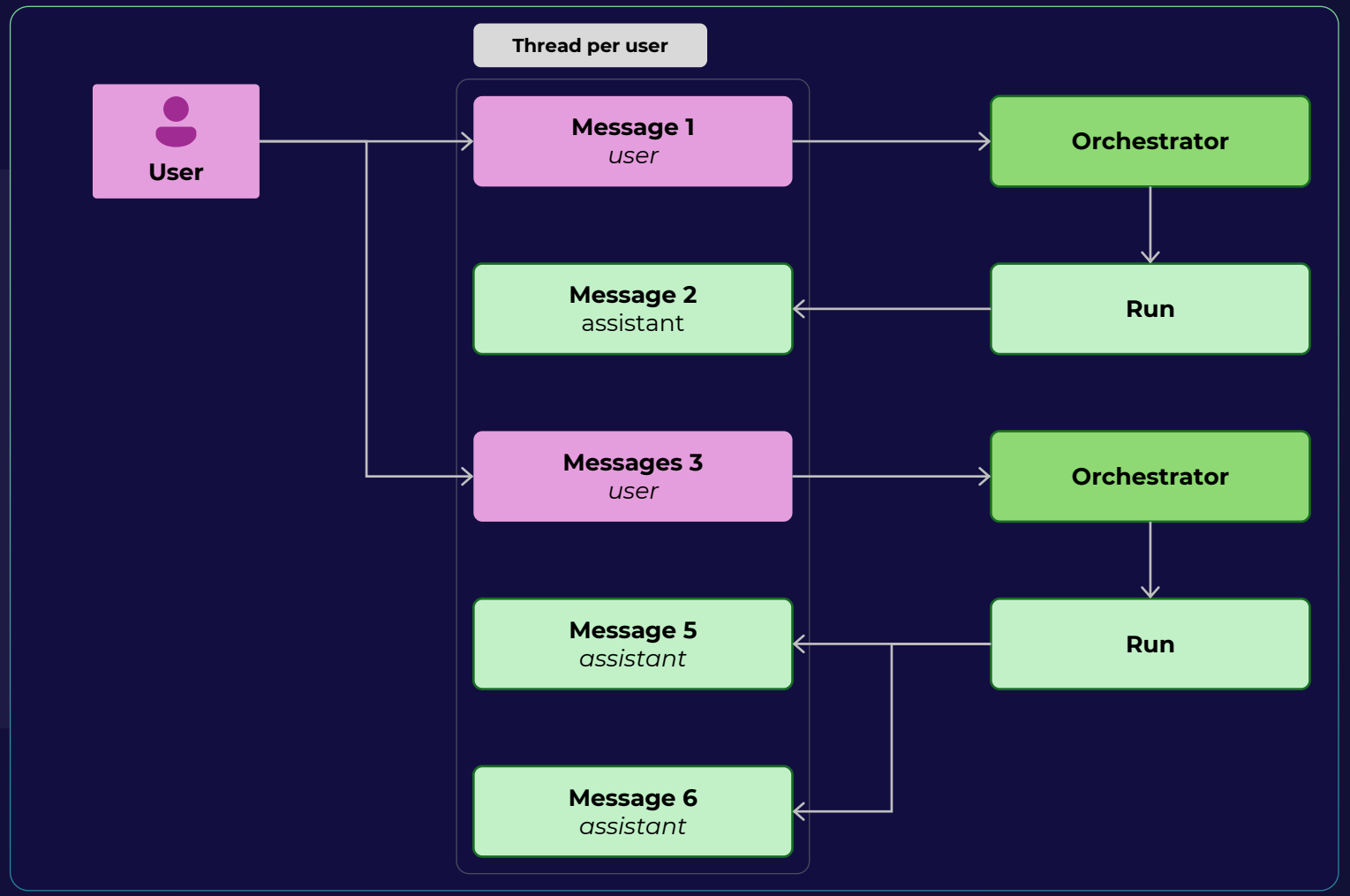
+ Add example | Import from .json | Download all as .json | Delete | Filter by keyword

Showing 1 items (0 selected)

Question	SQL query
Which Streets in Manchester have the highest crime rates?	SELECT Location AS StreetName, COUNT(*) AS NumberOfCrimes FROM [GreaterManchesterCrime].[greatermanchestercrime] -- note that we ALWAYS want to perform case insensitive searches
59/500 characters used	368/1000 characters used

Assistant APIs

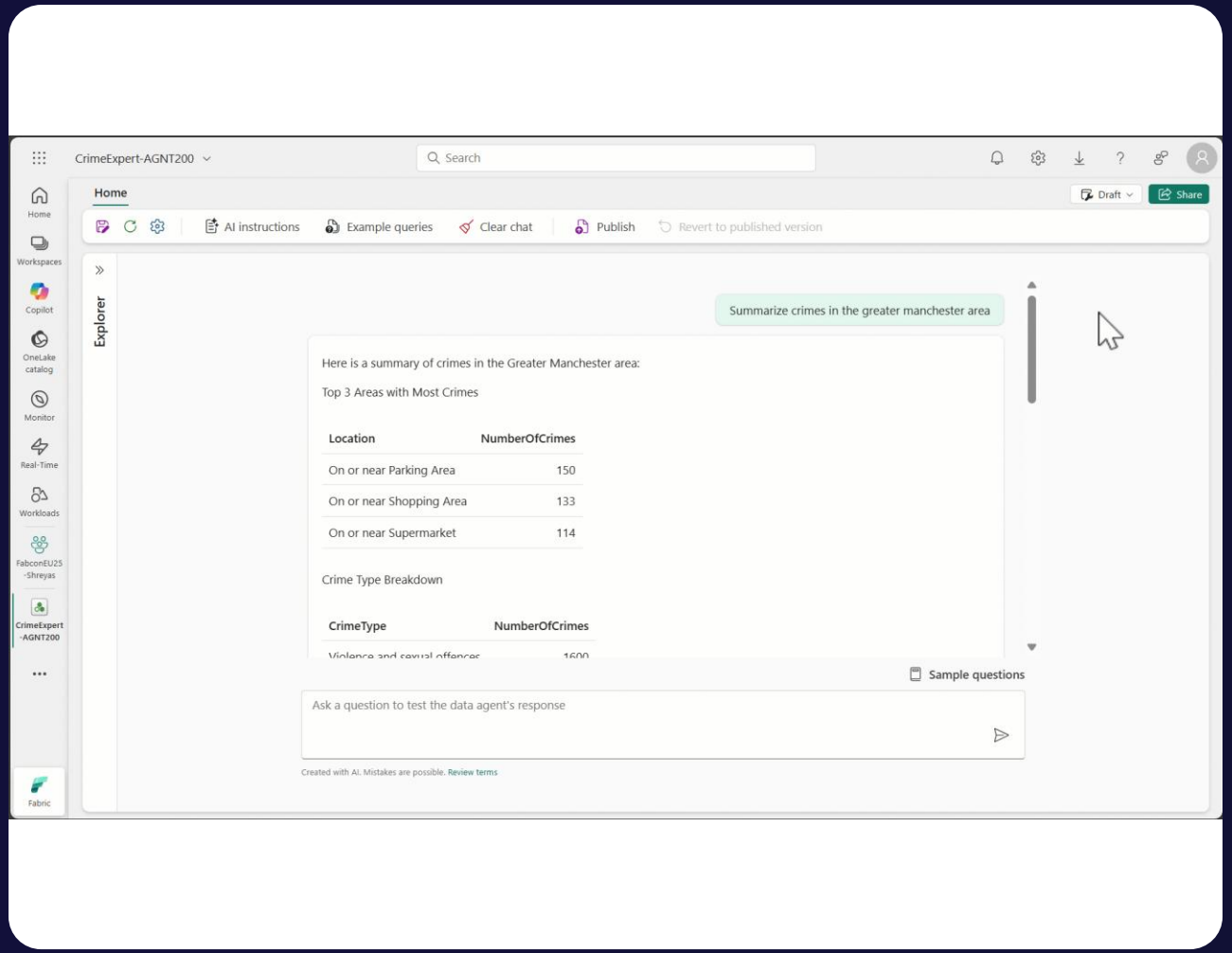
- **Messages**
- **Threads**
- **Runs**



Data Agent UX: Run Steps

Data Agent displays run steps:

- Tool that was chosen
- The query executed
- The query output



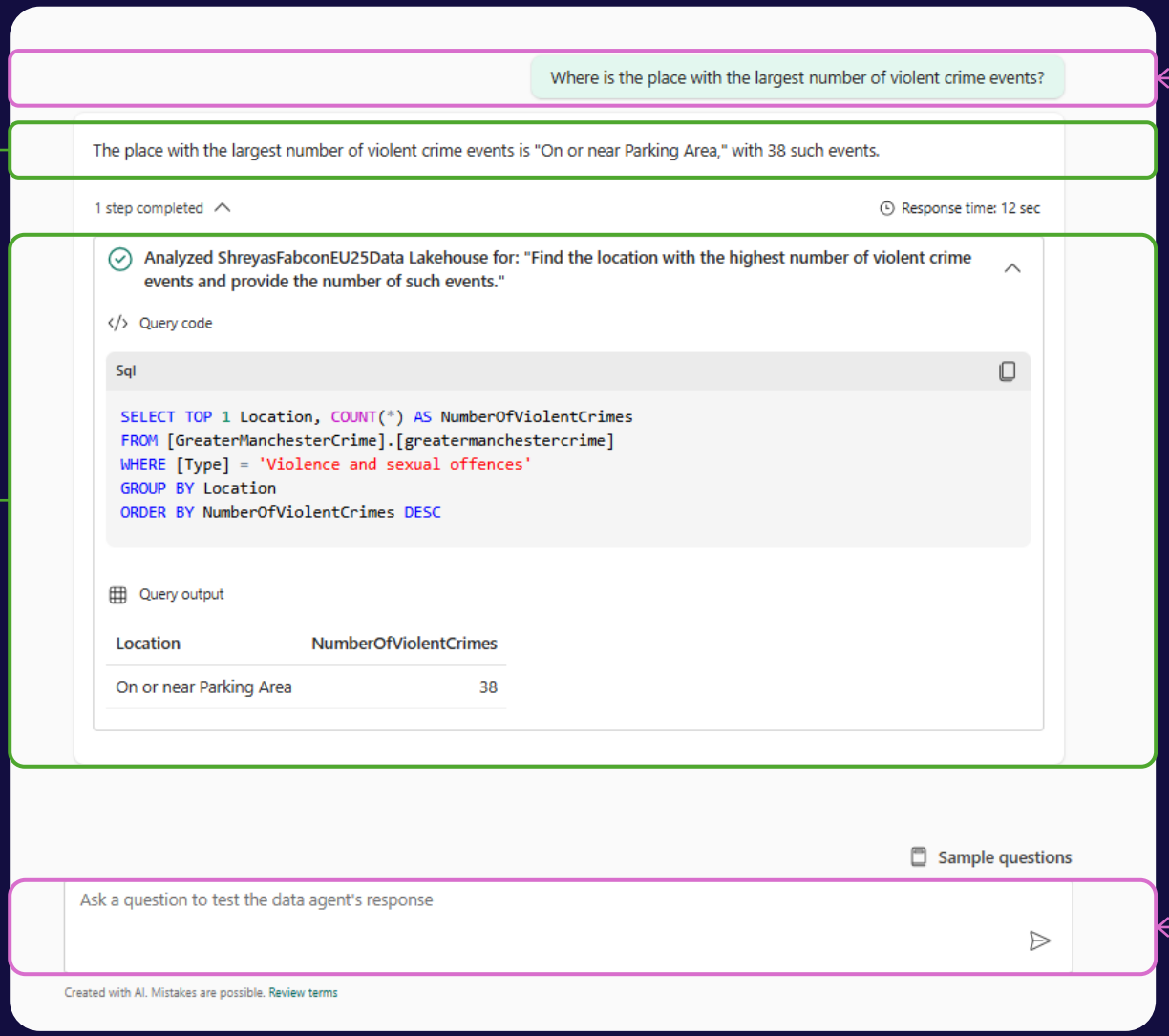
Thread

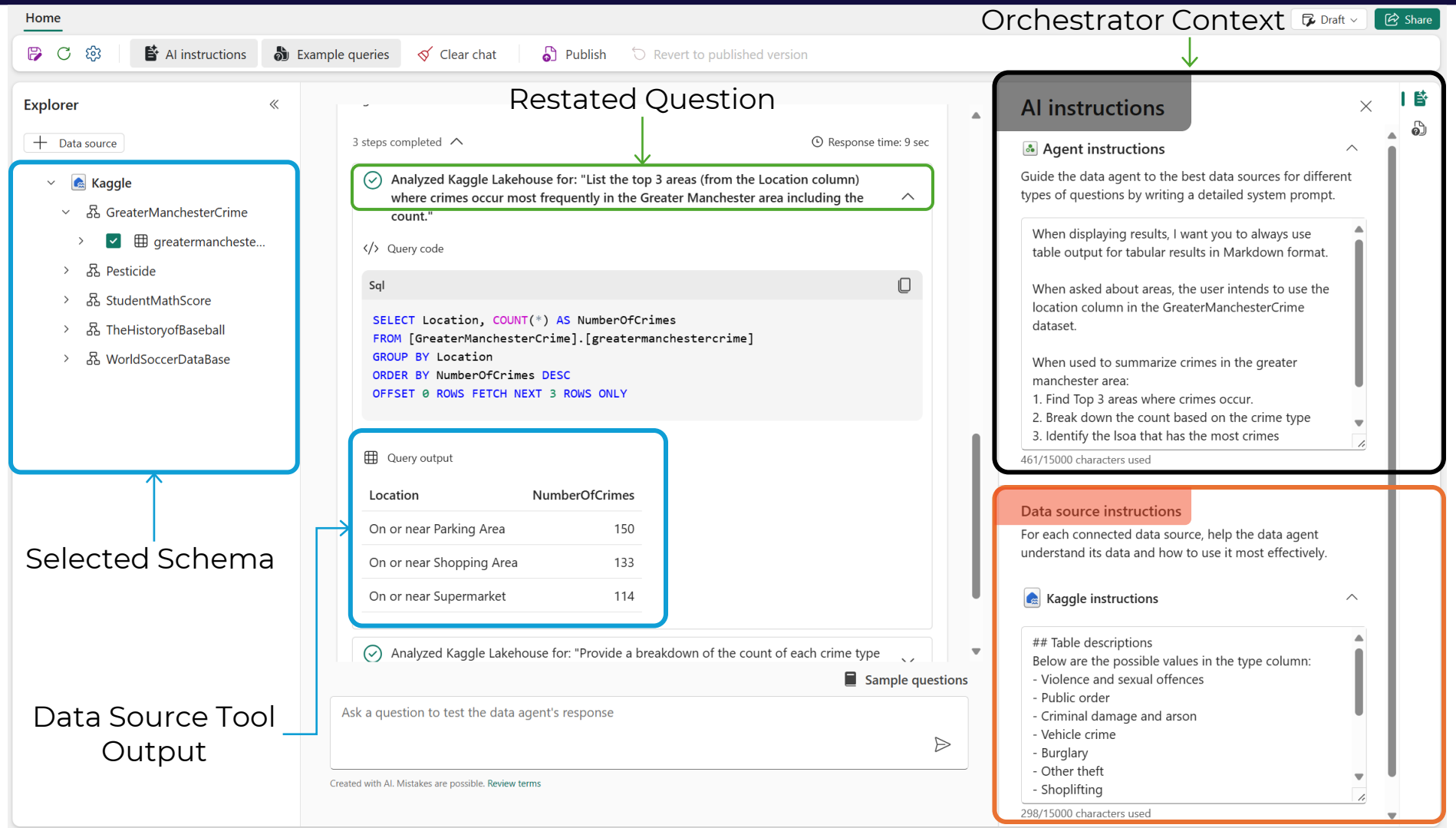
Message 1: Submitted by the user

Message 2: Assistant Generated Response based on Run steps

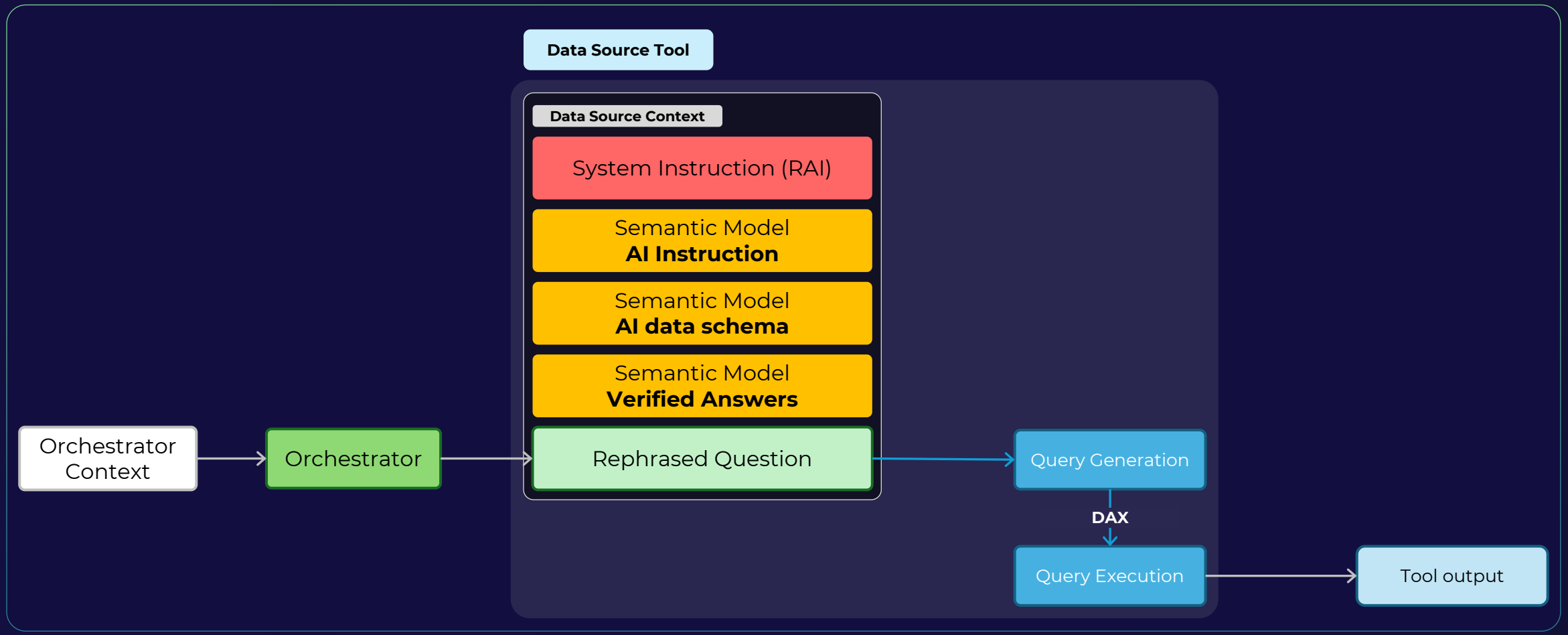
Run steps generated by Data Agent

User Input



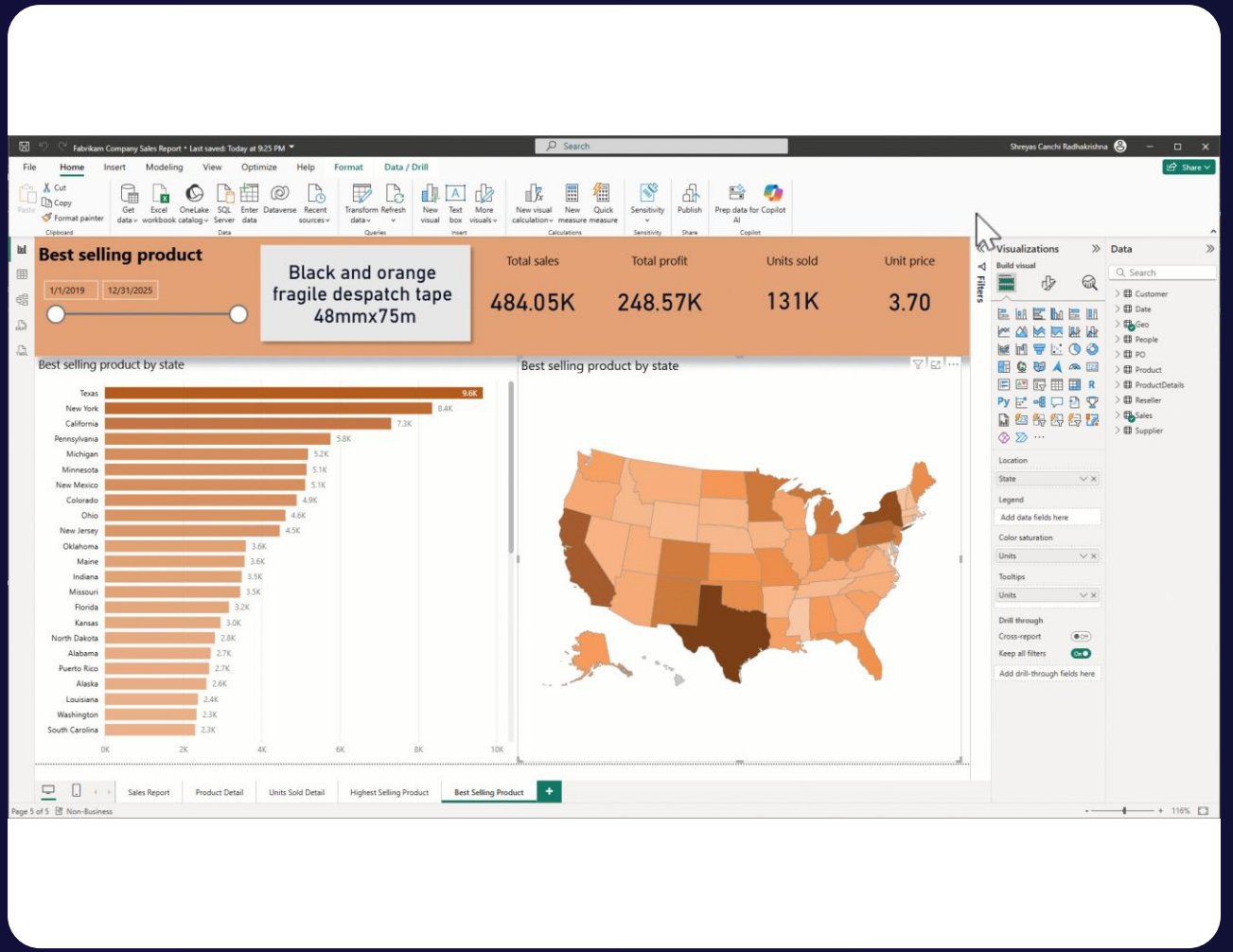

Home
Orchestrator Context
Draft
Share
AI instructions
Example queries
Clear chat
Publish
Revert to published version
Explorer
Data source
Kaggle
GreaterManchesterCrime
greatermancheste...
Pesticide
StudentMathScore
TheHistoryofBaseball
WorldSoccerDataBase
Restated Question
3 steps completed
Response time: 9 sec
Analyzed Kaggle Lakehouse for: "List the top 3 areas (from the Location column) where crimes occur most frequently in the Greater Manchester area including the count."
Query code
Sql
SELECT Location, COUNT(*) AS NumberOfCrimes
FROM [GreaterManchesterCrime].[greatermanchestercrime]
GROUP BY Location
ORDER BY NumberOfCrimes DESC
OFFSET 0 ROWS FETCH NEXT 3 ROWS ONLY
Query output
Location
NumberOfCrimes
On or near Parking Area
150
On or near Shopping Area
133
On or near Supermarket
114
Analyzed Kaggle Lakehouse for: "Provide a breakdown of the count of each crime type
Sample questions
Ask a question to test the data agent's response
Created with AI. Mistakes are possible. Review terms
AI instructions
Agent instructions
Guide the data agent to the best data sources for different types of questions by writing a detailed system prompt.
When displaying results, I want you to always use table output for tabular results in Markdown format.
When asked about areas, the user intends to use the location column in the GreaterManchesterCrime dataset.
When used to summarize crimes in the greater manchester area:
1. Find Top 3 areas where crimes occur.
2. Break down the count based on the crime type
3. Identify the lsoa that has the most crimes
461/15000 characters used
Data source instructions
For each connected data source, help the data agent understand its data and how to use it most effectively.
Kaggle instructions
Table descriptions
Below are the possible values in the type column:
- Violence and sexual offences
- Public order
- Criminal damage and arson
- Vehicle crime
- Burglary
- Other theft
- Shoplifting
298/15000 characters used
Selected Schema
Data Source Tool Output

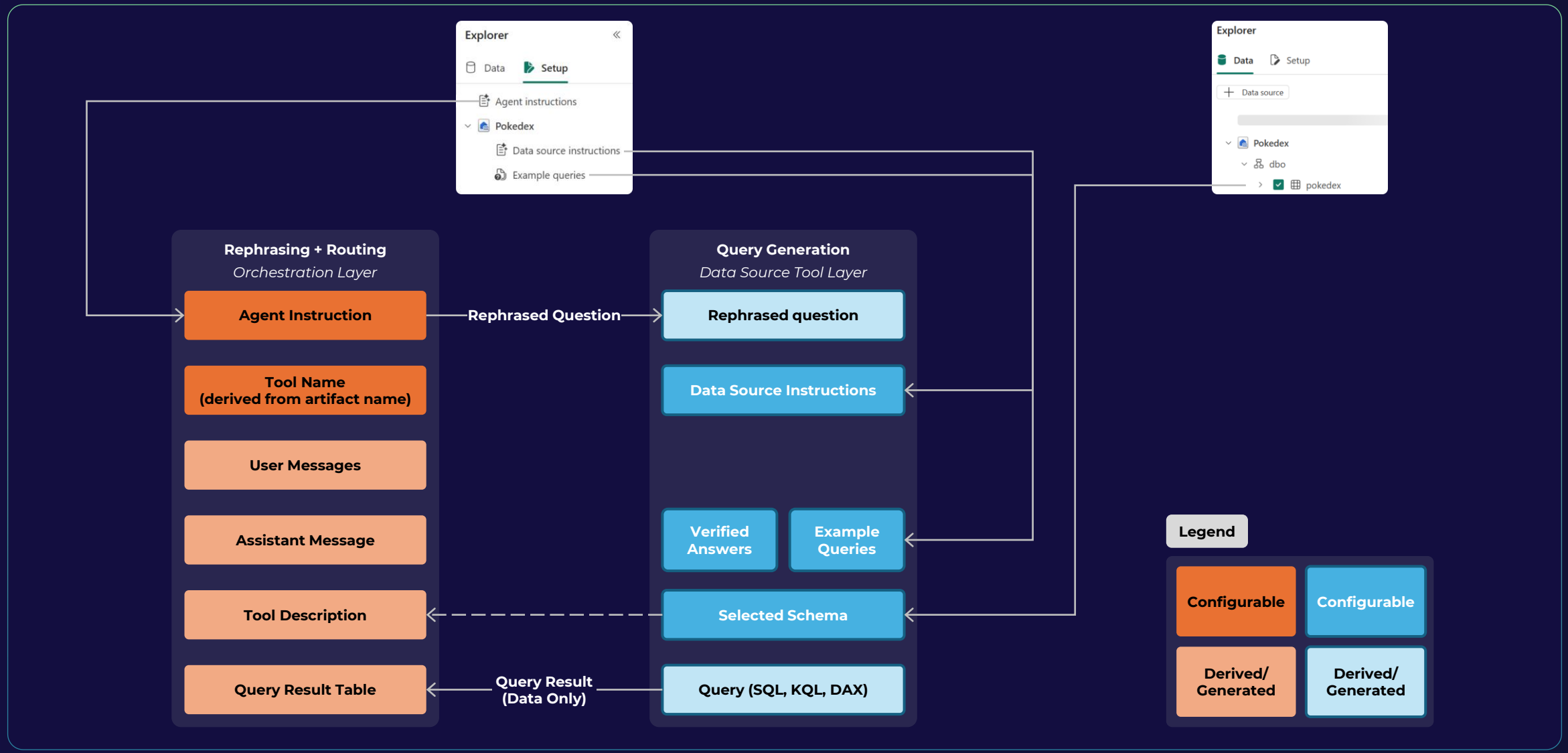
Data Source Tool: Semantic Model



Configure Semantic Models for AI

- Configure your Semantic Model in PBI Desktop
- Update Instructions, Verified Answers, and Schema Selection




Explorer
Data
Setup
Agent instructions
Pokedex
Data source instructions
Example queries
Explorer
Data
Setup
Data source
Pokedex
dbo
pokedex
Rephrasing + Routing
Orchestration Layer
Agent Instruction
Tool Name
(derived from artifact name)
User Messages
Assistant Message
Tool Description
Query Result Table
Rephrased Question
Query Result
(Data Only)
Query Generation
Data Source Tool Layer
Rephrased question
Data Source Instructions
Verified Answers
Example Queries
Selected Schema
Query (SQL, KQL, DAX)
Legend
Configurable
Configurable
Derived/ Generated
Derived/ Generated

Creator Configuration

Agent Instructions

Intended Use Cases:

- Specify which data sources to route for specific questions
- Guide how the agent interprets user intent

What to include:

- Map terminology to user intent
- Define steps for handling key questions
- Set response style and tone

Data Source Instructions

Intended Use Cases:

- Provide clarity on how to use each source
- Guide table selection for topics
- Ensure correct query logic

What to include:

- Descriptions of key tables & columns
- Recommended table usage for certain topics
- Example column values and filter patterns

Example Queries

Intended Use Cases:

- Data source instruction overflow (RAG)
- Reflect advanced logic
- Examples should highlight join patterns or expected values

What to include:

- Representative set of questions a consumer may ask

Q&A

Please submit questions at: https://aka.ms/dataagent-qa



Lab 2: Learn to Configure a Data Agent

Q&A

Please submit questions at: https://aka.ms/dataagent-qa



Data Agent SDK and Evaluation

Building a Data Agent is Iterative

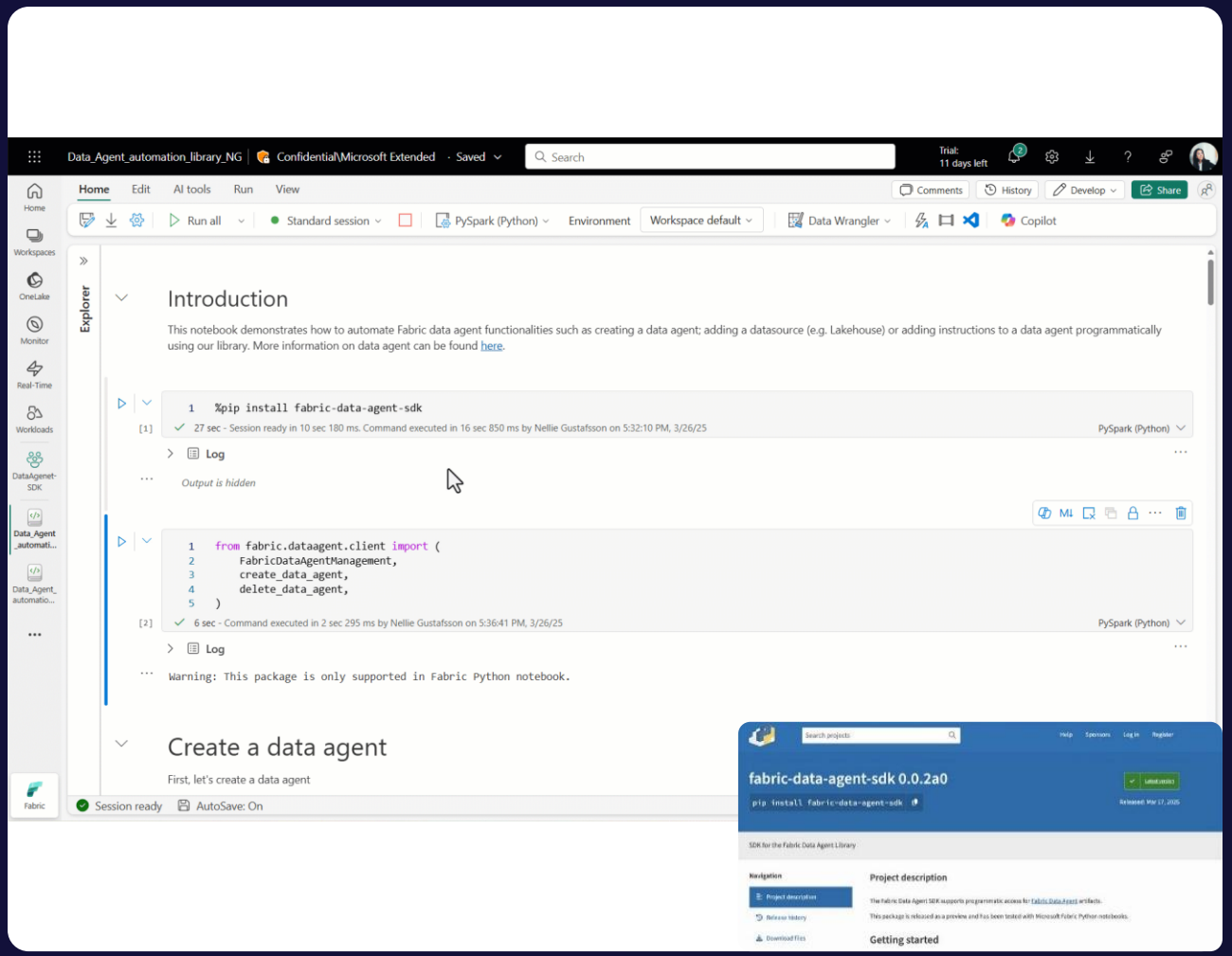


Fabric data agent SDK

Leverage the **Data Agent SDK** to invoke Fabric data agents from Fabric Notebooks

Create and configure data agents for automation and testing

Test and evaluate data agents against your ground truth datasets



Evaluate data agents

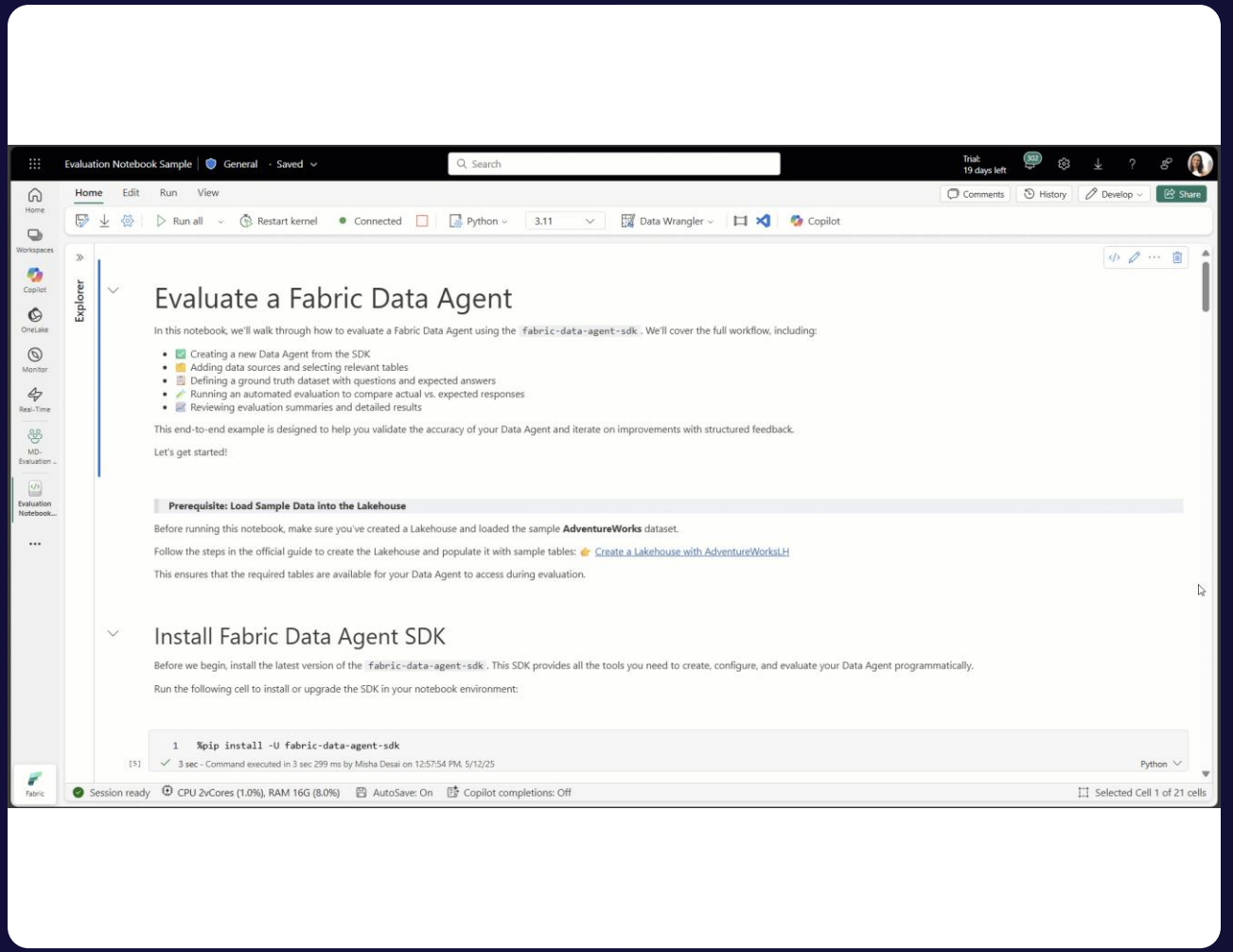
Native support for **data agent evaluation** through the Fabric data agent SDK

Run structured evaluations using Python from notebooks

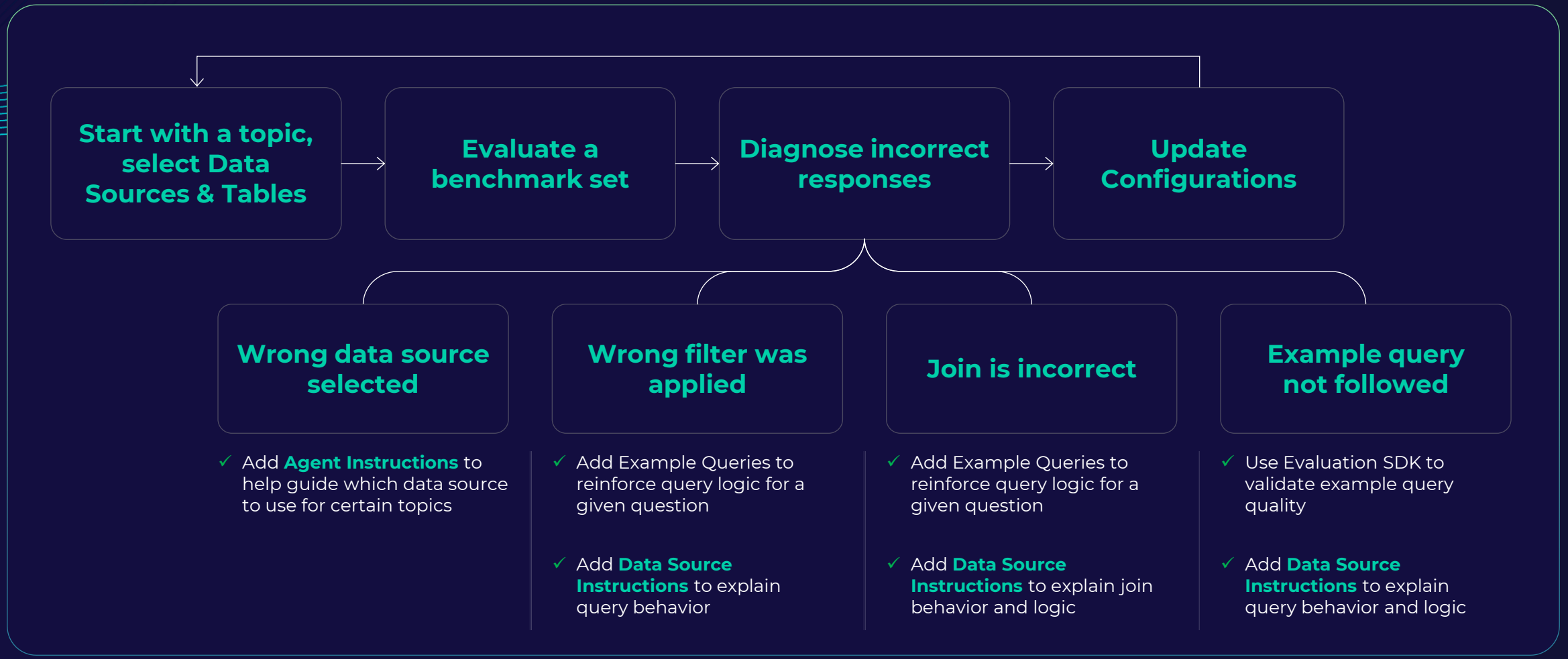
Validate accuracy before production deployment

Tune prompts for better performance

Benchmark improvements over time



Configuration Flow – Evaluation SDK



Generating a Benchmark

Understand the Consumer – what questions will they likely ask?

Clear and concise question + ground_truth pairs

Use aggregates rather than large outputs

- "List all product categories and their revenue in 2017" → "What are the top 3 Product Categories by Revenue"

Questions	Ground_truth
Give me the revenue from furniture related purchases	The revenue for furniture related purchases is **$1,143,364.98**
What was the revenue from health and beauty related purchases in 2017?	The Revenue from health and beauty related products purchased in 2017 is **$481,755.71**
Compute the median product price for category beauty and health.	The median price for health and beauty products is **79.9**

Evaluation Framework

Benchmark

Question	Ground_truth
What type of...	The type of cri...
...	..
...	...
...	...



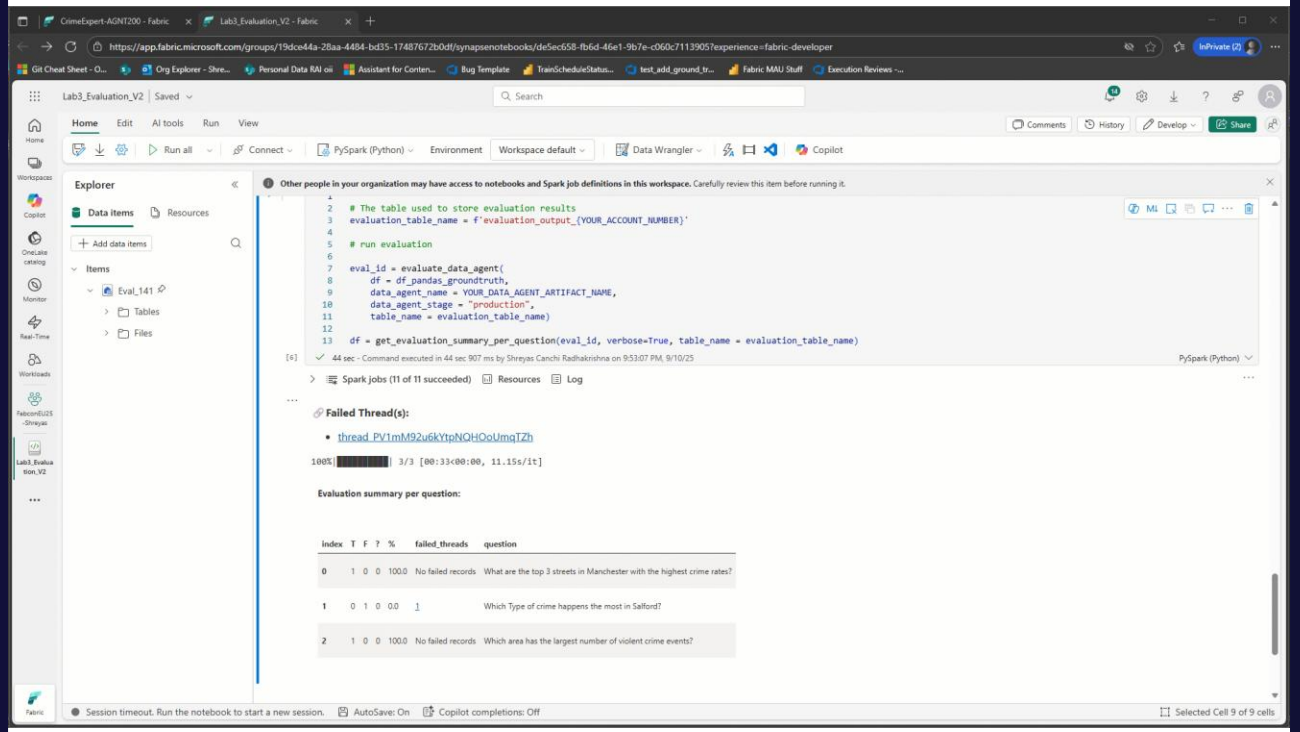
Evaluation Result

Result	Failed Thread	Question
Pass	N/A	What type of...
Fail	url	...
...	...	...
...	...	...

Diagnosing Incorrect Responses

Evaluation failures can be caused by multiple reasons

- Configs could improve
- Incorrect groundtruth
- Evaluator made a mistake
- Confusing Column names



Result Summary

🔗 **The result does not match the provided ground truth(s):**

- thread_77StYhqiKnIkPtKr7uB2G1SU
- thread_JIBrWUvJIt3fg3p7xdZtEp7V
- thread_xBaitmKoJyXFnuasoDWuV10R
- thread_1AMW6dPbszUq6vy8uOcj5Bkj

100%|██████████| 7/7 [00:52<00:00, 7.49s/it]

If a Thread has failed, you can find the link to the failed thread here.

Evaluation summary per question:

index	T	F	?	%	failed_threads	question
0	0	1	0	0.0	1	Compute the median product price for category beauty and health.
1	1	0	0	100.0	No failed records	Find the top city by the number of delivered orders.
2	0	1	0	0.0	1	Give me the revenue from furniture related purchases
3	0	1	0	0.0	1	Please provide the average, median, p95, and p99 for the delivery time of the City with the most number of deliveries.

index	question	expected_answer	actual_answer	evaluation_judgement	thread_url
0	Compute the median product price for category beauty and health.	The median price for health and beauty products is 79.9	No data was found for the median product price in the "beauty and health" (beleza_saude) category. This may mean there are no sales records with price data for this category, or that the price column is not available or named differently in the database. If you have another category or would like to adjust the search, please let me know!	False	thread_77StYhqiKnIkPtKr7uB2G1SU
1	Give me the revenue from furniture related purchases	The Revenue for furniture related purchases is 1143364.98000001	The closest matching product category to "furniture" is furniture_decor. The total revenue from furniture related purchases is $902,511.79. | Product Category | Total Revenue | |----------|----------:| | furniture_decor | $902,511.79 |	False	thread_JIBrWUvJIt3fg3p7xdZtEp7V
2	Please provide the average, median, p95, and p99 for the delivery time of the City with the most number of deliveries.	Zip Code: SAO PAULO Average: 7.97 Median: 7 P95: 19 P99: 31	Here are the delivery time statistics for the city with the most deliveries (SAO PAULO): | Statistic | Value (Delivery Days) | |----------|----------| | Average | 7 | | Median | 7 | | 95th Percentile | 19 | | 99th Percentile | 31 | These values are based on the Delivery Days for orders delivered in SAO PAULO.	False	thread_xBaitmKoJyXFnuasoDWuV10R
3	What was the revenue from health and beauty related purchases in 2017?	The Revenue from health and beauty related products purchased in 2017 is 550420.110000003	The total revenue from health and beauty related purchases in 2017 was 481,755.71.	False	thread_1AMW6dPbszUq6vy8uOcj5Bkj

Diagnosing the Issue

No data was found for the median product price in the "beauty and health" (beleza_saude) category. This may mean there are no sales records with price data for this category, or that the price column is not available or named differently in the database. If you have another category or would like to adjust the search, please let me know!

2 steps completed

Response time: 13 sec

Analyzed ecommerce_lakehouse Lakehouse for: "Compute the median product price for the product category 'beleza_saude' using the relevant product price or sales columns."

ıkehouse for: "Find the exact product category key(s) in fda.Products that match or ıı'."

ıkehouse for: "Compute the median product price for the product category roduct price or sales columns."

Query code

Sql

```
SELECT TOP 1
    P.[Product Category English] AS ProductCategory,
    PERCENTILE_CONT(0.5)
        WITHIN GROUP (ORDER BY S.Price)
        OVER(PARTITION BY P.[Product Category English]) AS MedianPrice
FROM
    [fda].[Sales] S
INNER JOIN
    [fda].[Products] P ON S.[Product ID] = P.[Product ID]
WHERE
    P.[Product Category English] = 'beleza_saude' AND
    S.Price IS NOT NULL
```

Query output

Failed to execute step (RAID: 37a1be78-a37a-44c7-8129-0be1c1804e4a). Error: No data found after query execution.

Incorrect Filter due to rephrased question

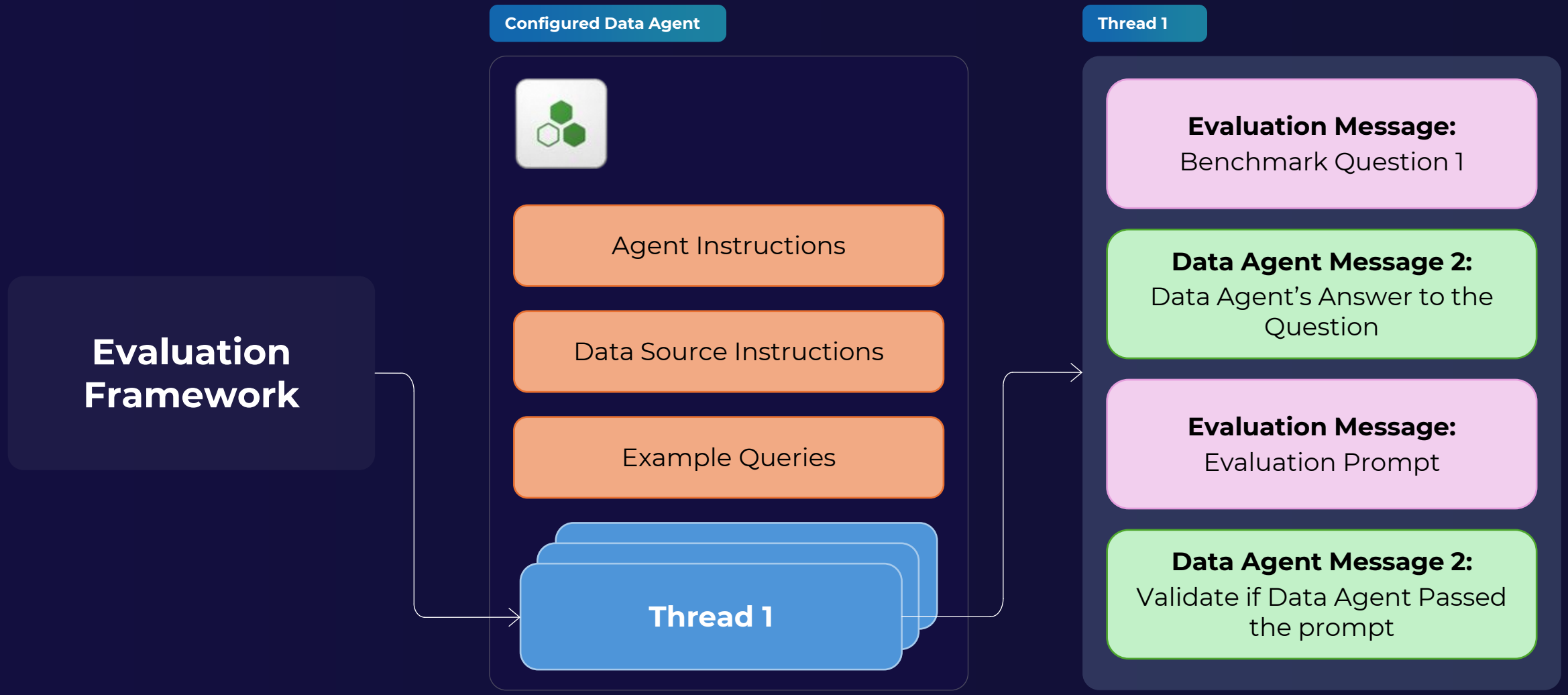
Diagnosing the Issue

	Product ID	Product C...	Product Catego...	Product Name L...	Product Descrip...
1	e3e020af31d4d89d2...	beleza_saude	health_beauty	58	1987
2	c5d8079278e912d7...	beleza_saude	health_beauty	60	330
3	36555a2f528d7b2a2...	beleza_saude	health_beauty	51	1069
4	e586ebb6022265ae...	beleza_saude	health_beauty	40	1469
5	75b4372e69a42f8ae...	beleza_saude	health_beauty	22	1313
6	3569d4374a919941...	beleza_saude	health_beauty	37	1168

The Column Data Agent Used (Product Category)

The Column I wanted data agent to use (Product Category English)

Evaluation Framework



Evaluation Prompt – LLM as a Judge

Given the following query and ground truth, please determine if the most recent answer is equivalent or satisfies the ground truth. If they are numerically and semantically equivalent or satisfy (even with reasonable rounding), respond with “Yes”. If they clearly differ, respond with “No”. If it is ambiguous or unclear, respond with “Unclear”. Return only one word: Yes, No, or Unclear.

Query: {query}

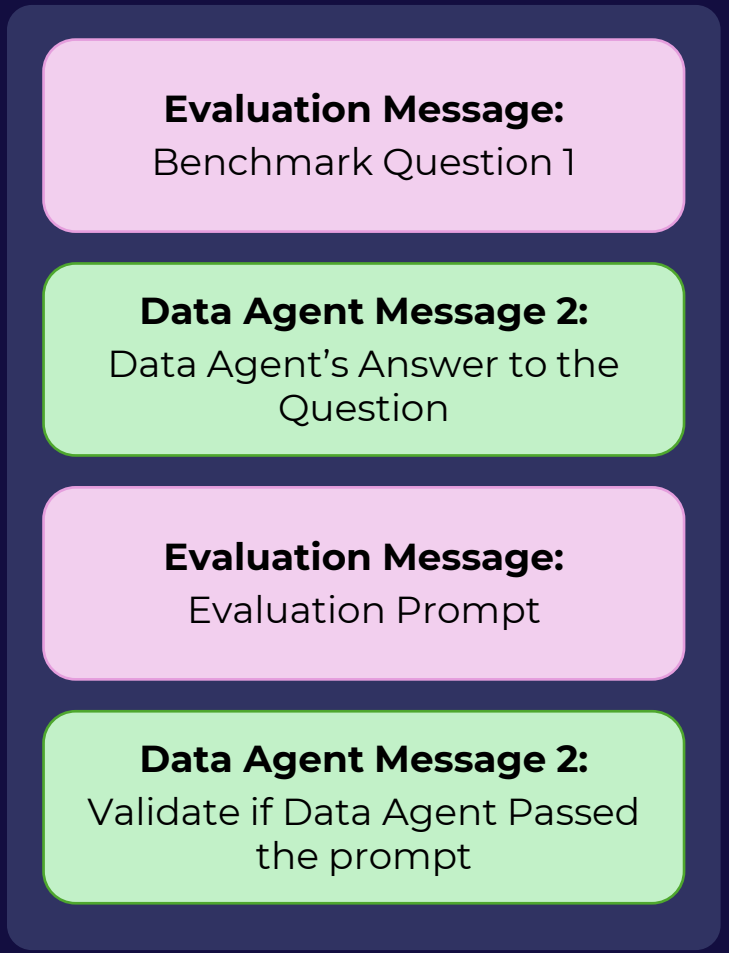
Ground Truth: {ground_truth}

Evaluation is One-Shot

We evaluate on a single question, and expect the response to be correct.

Talking to Data Agent allows you to talk to the Agent. This means:

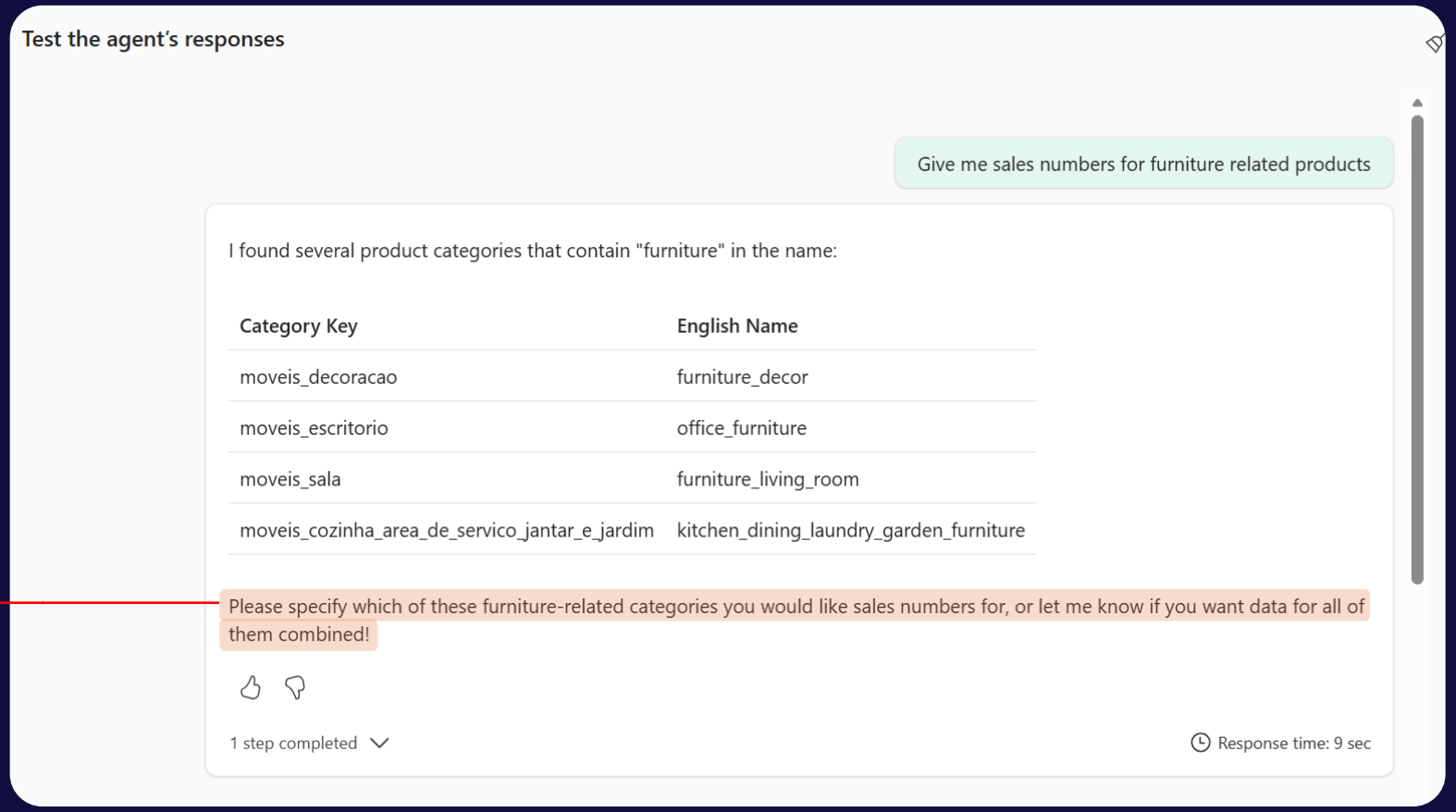
- Provide more context
- Clarify Assumptions
- Build on top of previous results



Evaluation is One-Shot

There are multiple ‘furniture’ related products.

A User may provide additional input to the Data Agent.



Evaluation is One-Shot

Question in the Evaluation:

"Give me the revenue from furniture related purchases"

🔗 **The result does not match the provided ground truth(s):**

- thread_GP6RXKhNZbEBbuofN90ABHfX

```
100%|██████████| 7/7 [01:43<00:00, 14.75s/it]
```

Evaluation summary per question:

index	T	F	?	%	failed_threads	question
0	1	0	0	100.0	No failed records	Compute the median product price for category beauty and health.
1	1	0	0	100.0	No failed records	Find the top city by the number of delivered orders.
2	1	0	0	100.0	No failed records	Give me the revenue from furniture related purchases

Update Configurations

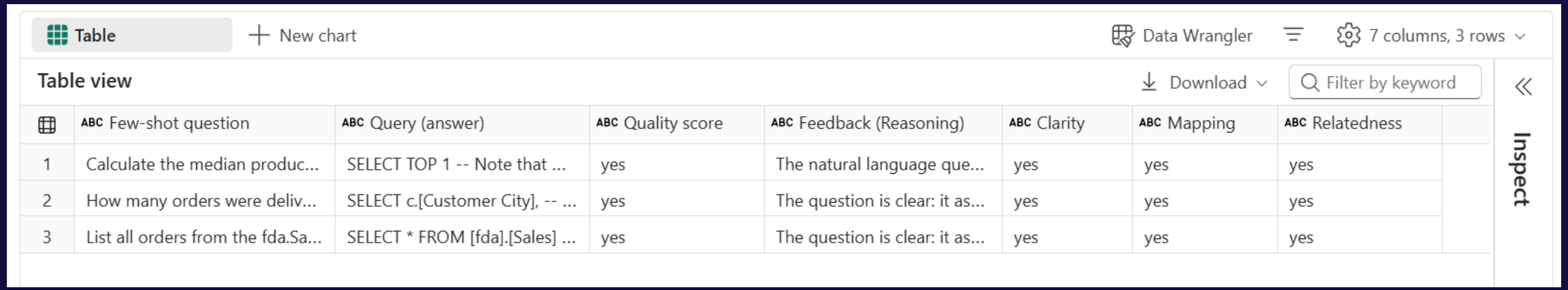
Wrong data source selected	Wrong filter was applied	Join is incorrect	Example query not followed
✓ Add **Agent Instructions** to help guide which data source to use for certain topics	✓ Add Example Queries to reinforce query logic for a given question ✓ Add **Data Source Instructions** to explain query behavior	✓ Add Example Queries to reinforce query logic for a given question ✓ Add **Data Source Instructions** to explain join behavior and logic	✓ Use Evaluation SDK to validate example query quality ✓ Add **Data Source Instructions** to explain query behavior and logic

Evaluation Few-Shot Quality

Leverage the Data Agent SDK to Evaluate Few-Shot Quality. Across 3 Metrics:

1. Clarity
2. Mapping
3. Relationships

Note: This does not apply to Semantic Models for the labs



	Few-shot question	Query (answer)	Quality score	Feedback (Reasoning)	Clarity	Mapping	Relatedness
1	Calculate the median produc...	SELECT TOP 1 -- Note that ...	yes	The natural language que...	yes	yes	yes
2	How many orders were deliv...	SELECT c.[Customer City], -- ...	yes	The question is clear: it as...	yes	yes	yes
3	List all orders from the fda.Sa...	SELECT * FROM [fda].[Sales] ...	yes	The question is clear: it as...	yes	yes	yes

Evaluation Few-Shot Quality

Clarity

- Measures whether the natural language question is **clear and unambiguous.**
- Questions should be specific, include necessary metrics, timeframes, and filters, and avoid vague or multi-intent phrasing.

Relatedness

- Evaluates how closely the SQL query matches the **intent of the natural language question**.
- The SQL should return the correct metric, apply the proper filters, and match the requested granularity.

Mapping

- Checks whether **all literals in the natural language question appear in the SQL query**.
- Every number, date, or category mentioned in the question should be explicitly represented in the SQL.

Q&A

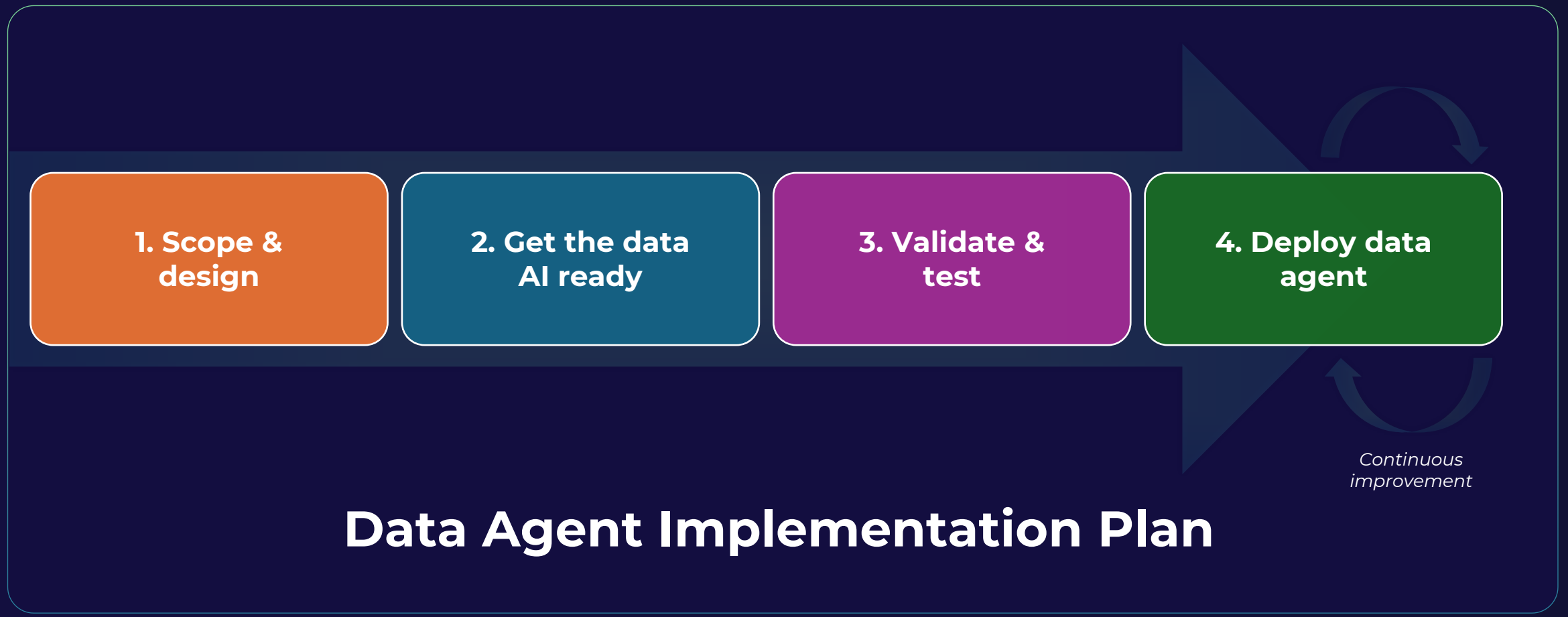
Please submit questions at: https://aka.ms/dataagent-qa



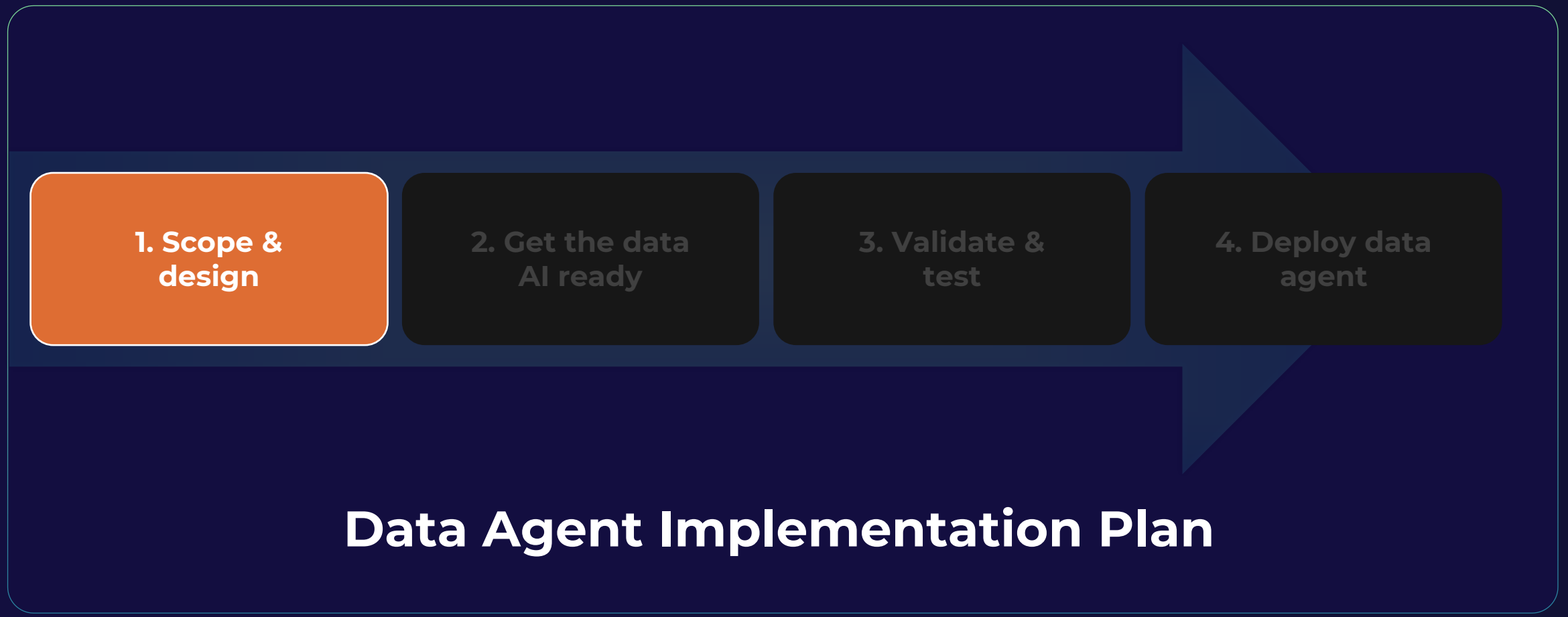
Lab 3: Evaluation

Blueprint for Microsoft Fabric Data Agent Success

Planning a Successful Data Agent Implementation



Planning a Successful Data Agent Implementation



Why Scoping Matters

Why	Best practice	Avoid
• Agents with laser-focused mission deliver value • Specialized agents tie to a business outcome • General-purpose agents with vague objectives fall short	• Keep the agent's scope as narrow as possible while still delivering value • Minimize data source scope	• Overloading with too much data by maximizing data sources • Over-engineering the agent instructions

Scoping checklist

Who are the primary users of the data agent?	What are the data agent's main functions?	What are the top questions users are expected to ask the data agent?	What topics are out of scope?	Are the goals of the data agent achievable with the data currently available in Fabric?
Who will use this agent, and for what exact tasks or decisions? For example, is it for analysts querying sales data or support staff searching knowledge? Define the user persona.	List the specific things the agent should do (e.g. "answer inventory levels by store", "summarize customer feedback from logs").	This will help clarify the types of queries the agent needs to support. Use these questions to evaluate whether any are too broad or not suitable for the data agent and provide feedback to refine them, so they are clearly tied to business outcomes.	Clearly note what the agent will not handle. For instance, a sales data agent won't perform HR queries. This manages expectations and prevents scope creep.	Verify that the questions in scope are answerable with available data in Fabric. Avoid "why" or open-ended questions the agent can't handle, focus on what the data can directly answer.

Examples

General purpose data agent

What are our **top-selling** products this week?

What are the top **complaint** categories in LATAM?

Which **campaigns** had the highest ROI last quarter?

What's our **gross margin** trend YoY?

Which warehouses are running low on **stock**?

PERFORMANCE ISSUES

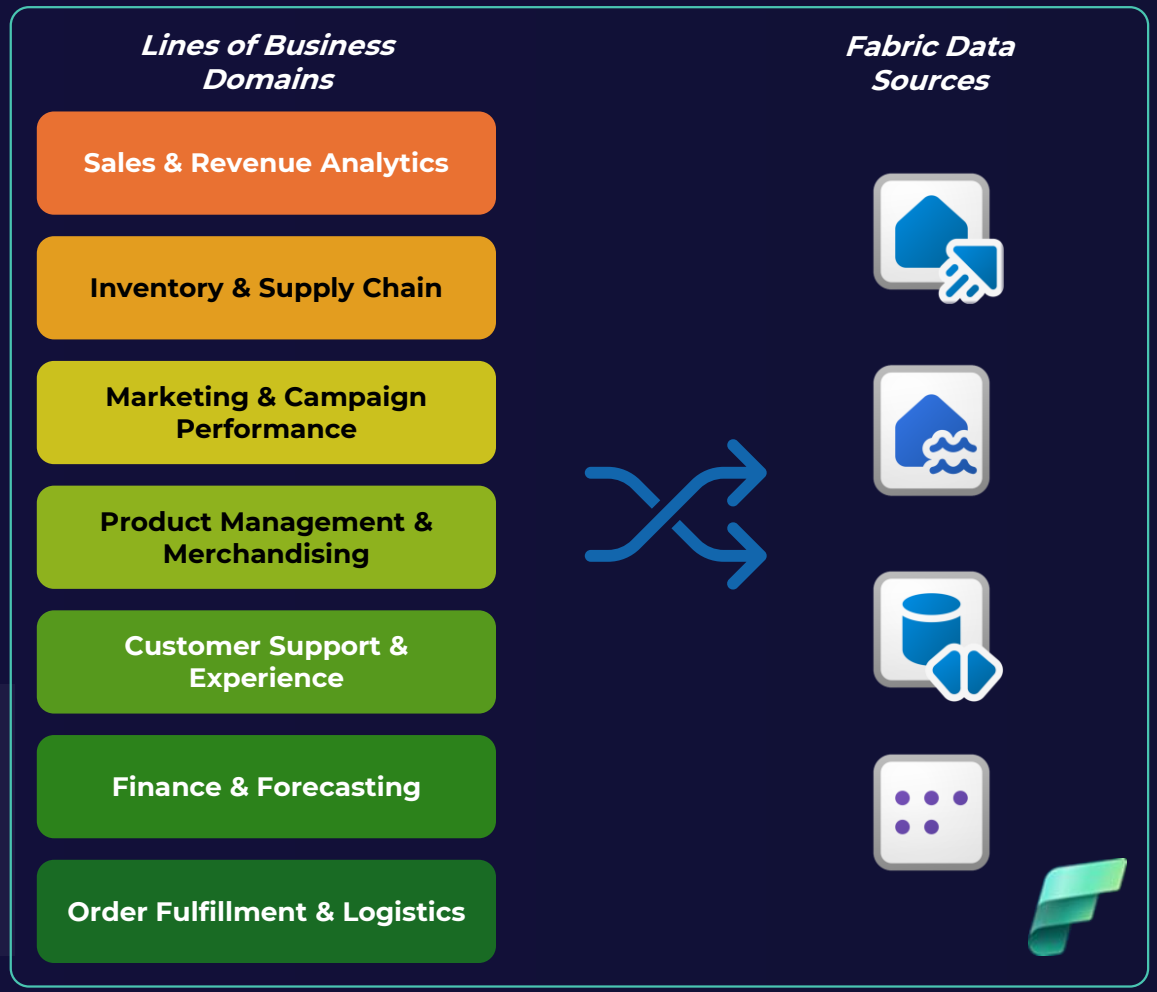
HARD TO MAINTAIN

NO CLEAR OWNERSHIP

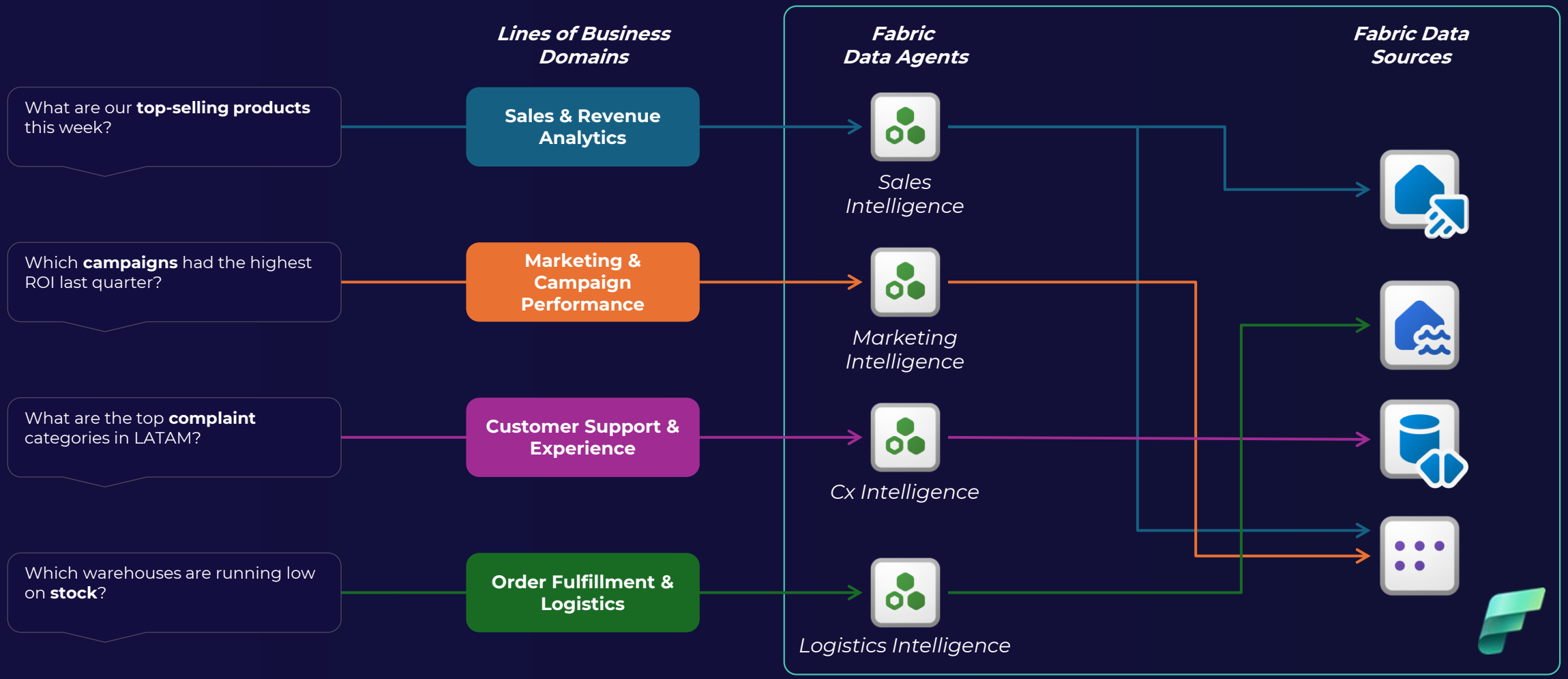
CONFUSED LOGIC

LOW TRUST

"Everything agent"

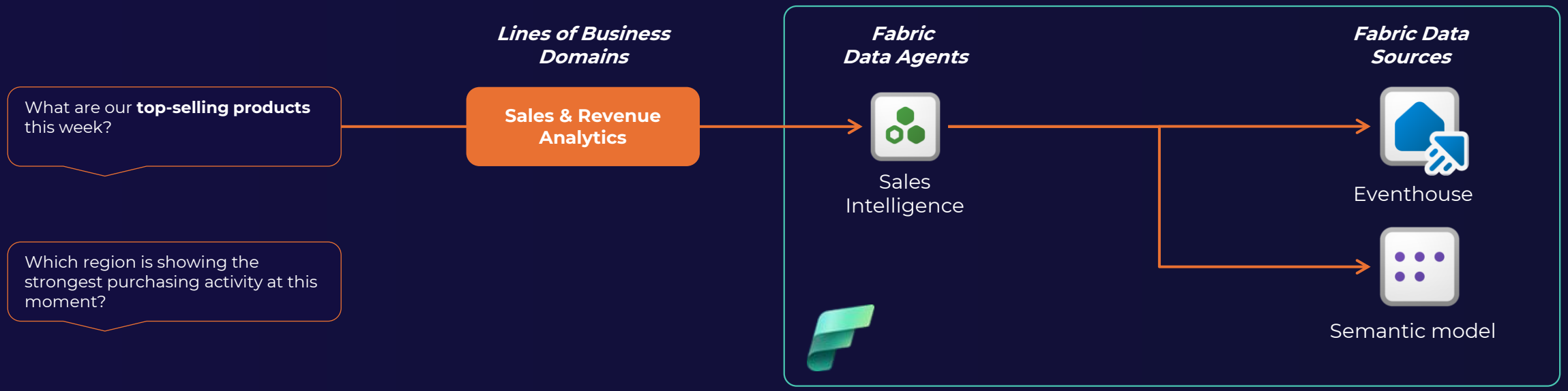


Domain-specific data agents



Real-time Sales Intelligence

Example domain-specific data agent



Real-Time Sales Intelligence Data Agent

Purpose: Empower sales analysts and regional managers to query real-time order data and monitor sales performance across geographies and customer segments.

Data Sources: Eventhouse (real–time order events), Power BI semantic model for sales KPIs.

Scope: Only sales-related queries.

RevOps Intelligence

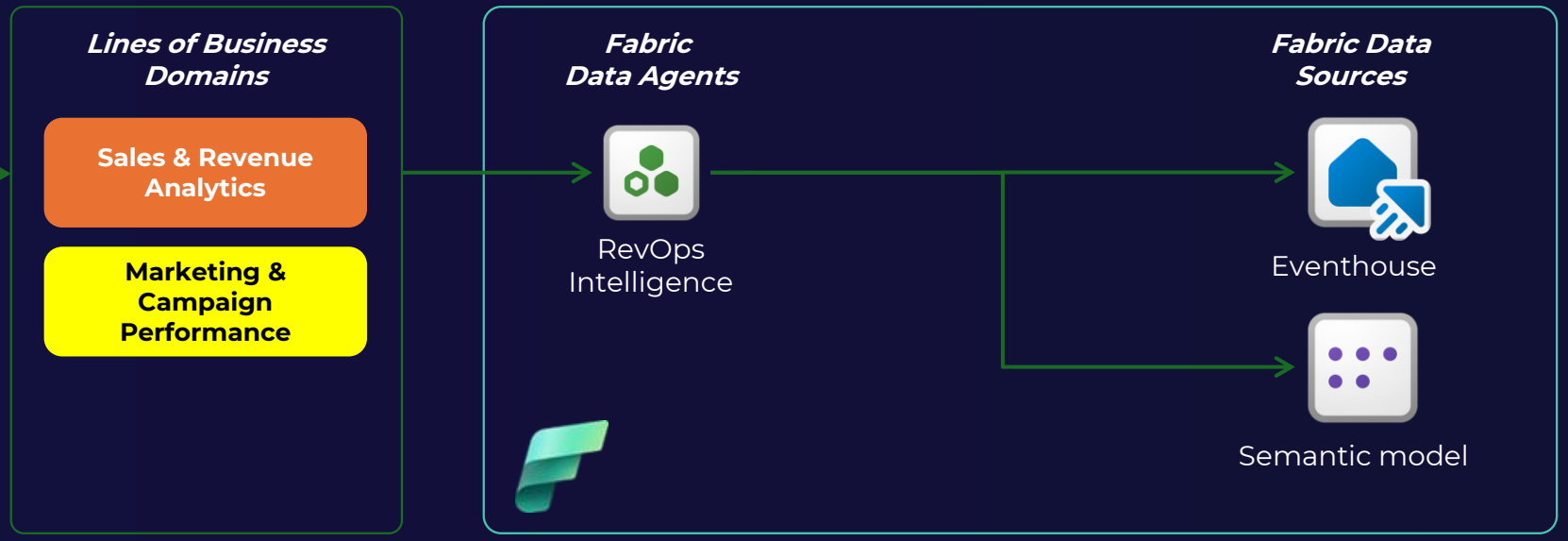
Example domain-specific data agent

How does paid **social media performance** correlate with **sales uplift** in the beauty category?

What's the **ROI** of our Q1 influencer campaigns across APAC?

Which **marketing channels** are driving the most high-value customer purchases today?

Which campaigns are generating the **highest revenue** per dollar spent this week?



Sales + Marketing Intelligence Data Agent

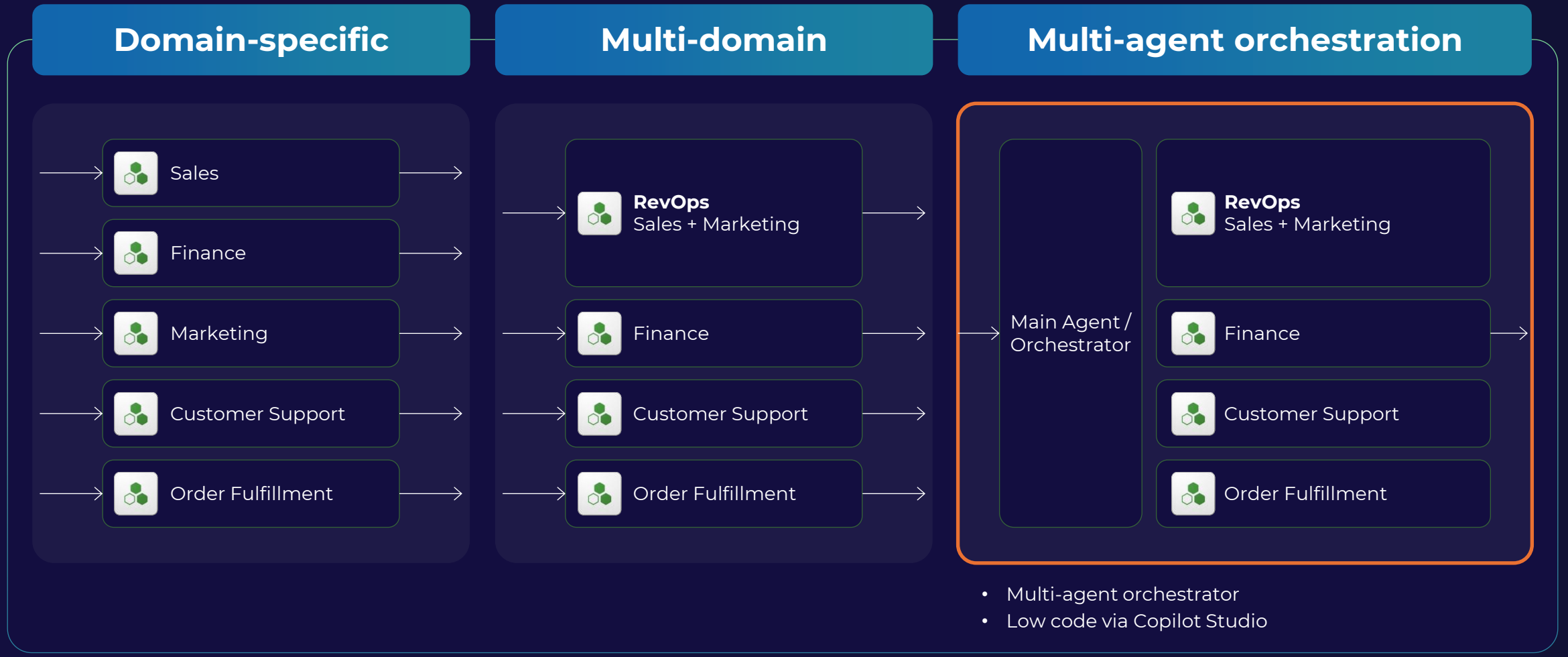
Purpose: Empower sales analysts with a unified view of campaign performance and its direct impact on sales.

Data Sources: Eventhouse (real-time order events), Power BI semantic model for campaign data and sales KPIs.

Scope: Focused exclusively on marketing and sales performance. Designed for campaign ROI analysis.

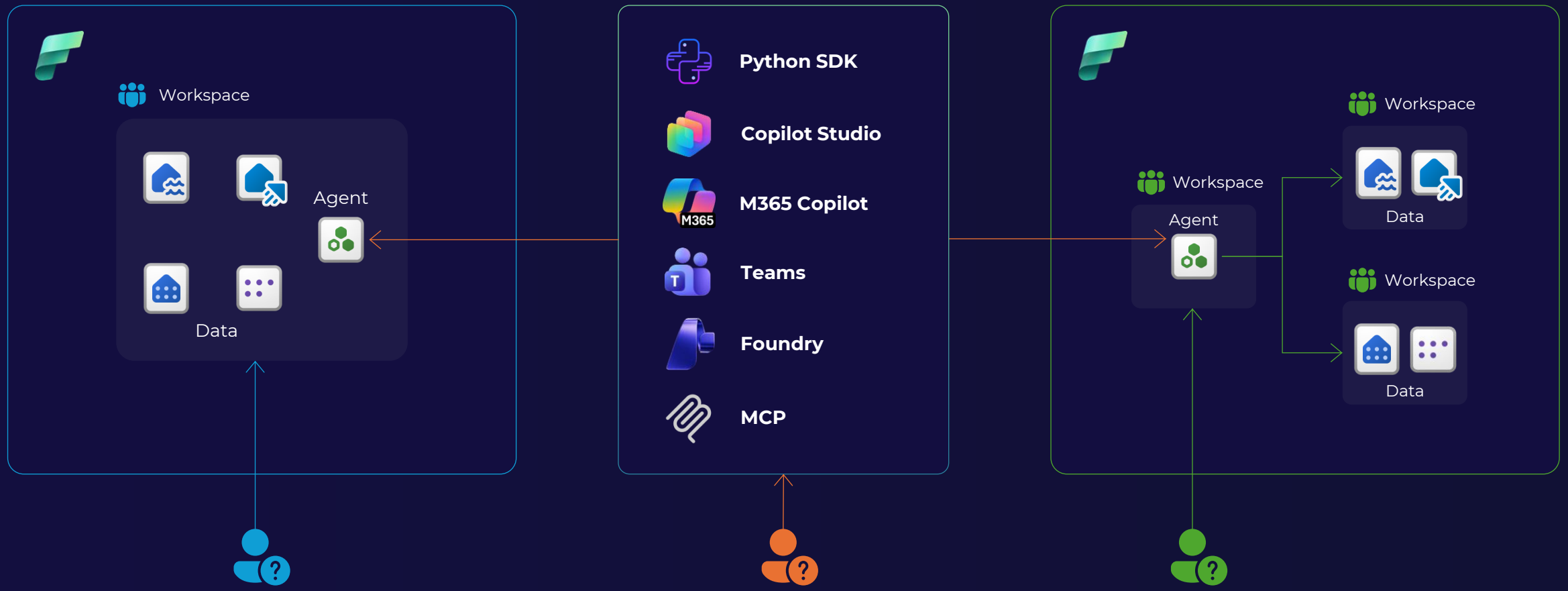
Patterns

Design patterns

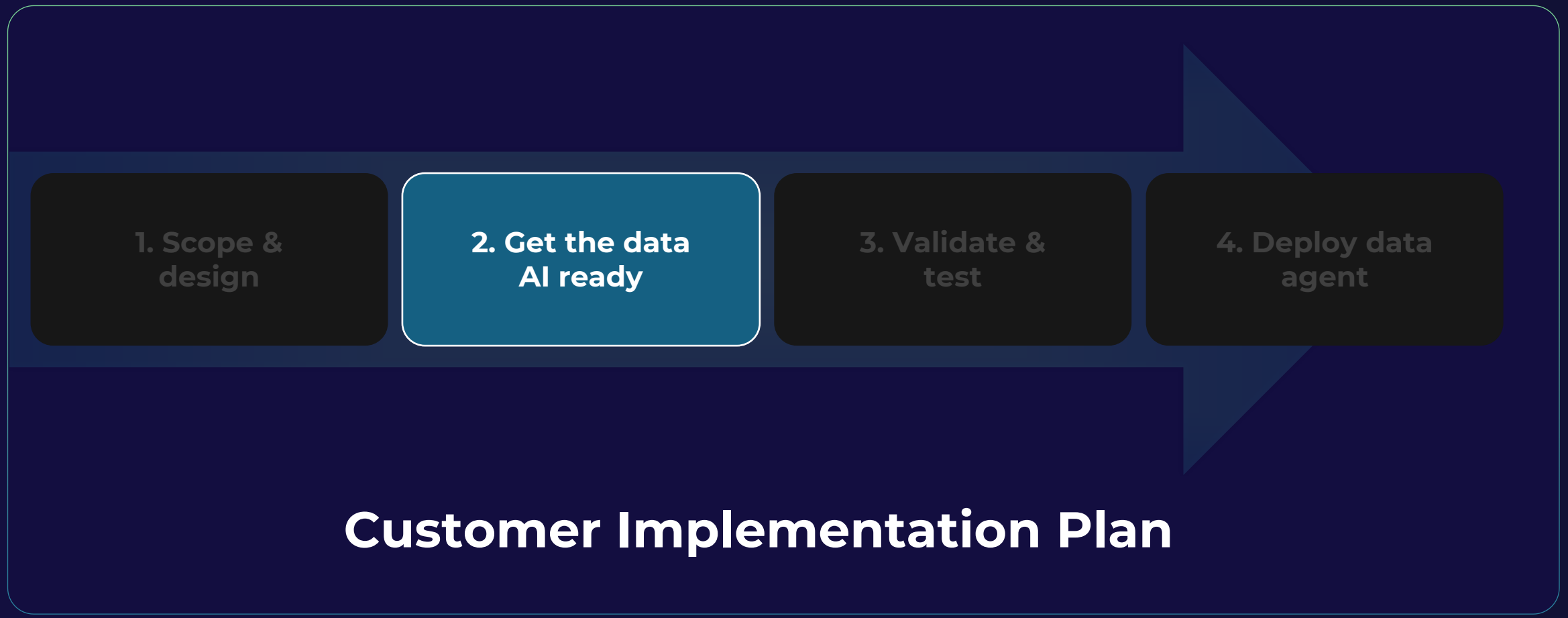


- Multi-agent orchestrator
- Low code via Copilot Studio

Access patterns



Planning a Successful Data Agent Implementation



Assess Data Availability



Data readiness checklist

List data sources and assess gaps

Inventory the structured data relevant to the agent's scope. This includes Lakehouses, Warehouses, Eventhouses and Power BI datasets (semantic models).

For each key question in scope, ensure there is a data source/field that can answer it. If you have a gap, decide how to address it. Can it be derived? Or out of scope?

Check data freshness

How up-to-date must the data be for this agent? Identify data that needs frequent refresh (e.g. daily sales) and ensure those **pipelines** are in place.

Review data quality

Profile the data for nulls, duplicates, or obvious errors. Aim for high-quality data in key fields. Think of the agent as unforgiving, it will surface whatever data is there, right or wrong.

Also, poor **metadata** makes it hard for the agent to find the right info. Note which data sources have good business-friendly schema, and which may need cleanup or aliasing.

Evaluate volume and performance

If the data source is extremely large (billions of rows), confirm that it's partitioned or indexed for performance.

Move data if needed

Ensure needed data is in (or connected to) Fabric OneLake. Fabric Data Agents can only directly query Fabric-integrated data sources. If some data sits elsewhere plan a migration or real-time integration before including it.

Frequent questions on data source best practices



Which data source is the best for my data agent?

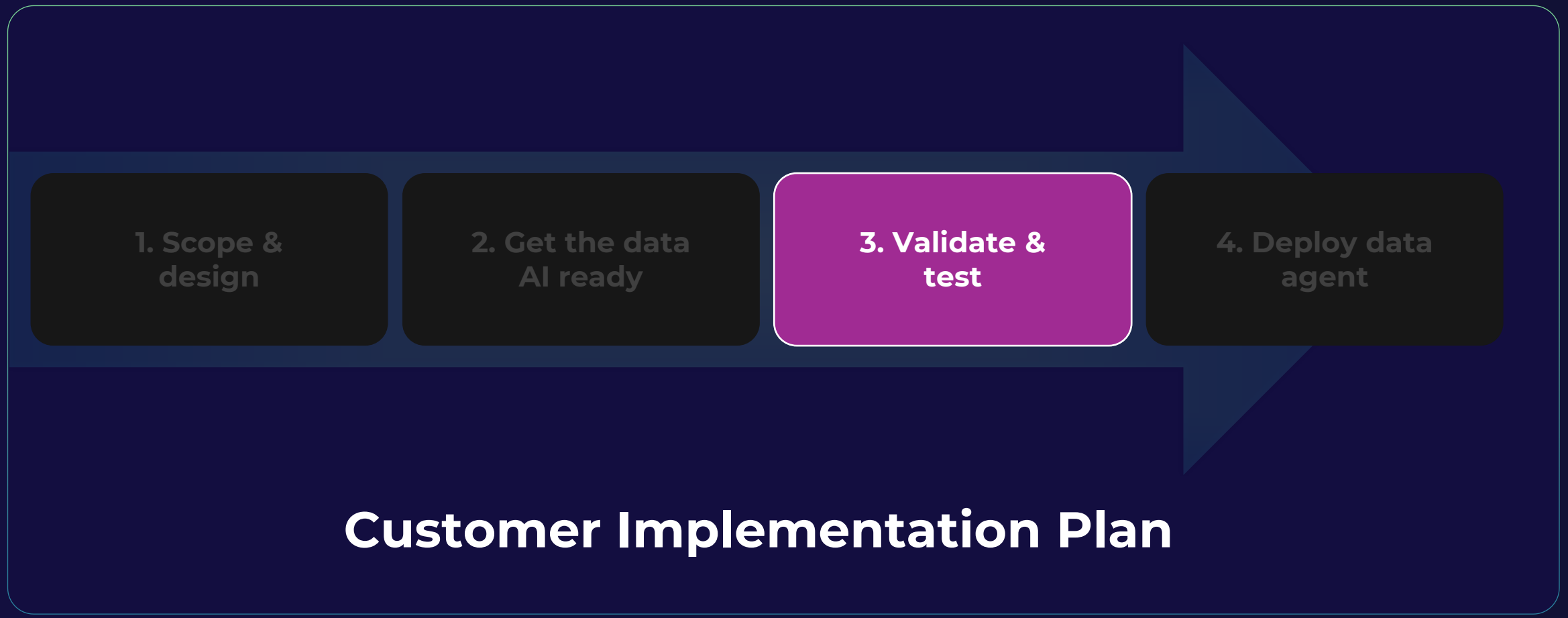


What does it mean to have AI-ready data?



How detailed should our AI instructions really be?

Planning a Successful Data Agent Implementation



Test & optimize

Test:

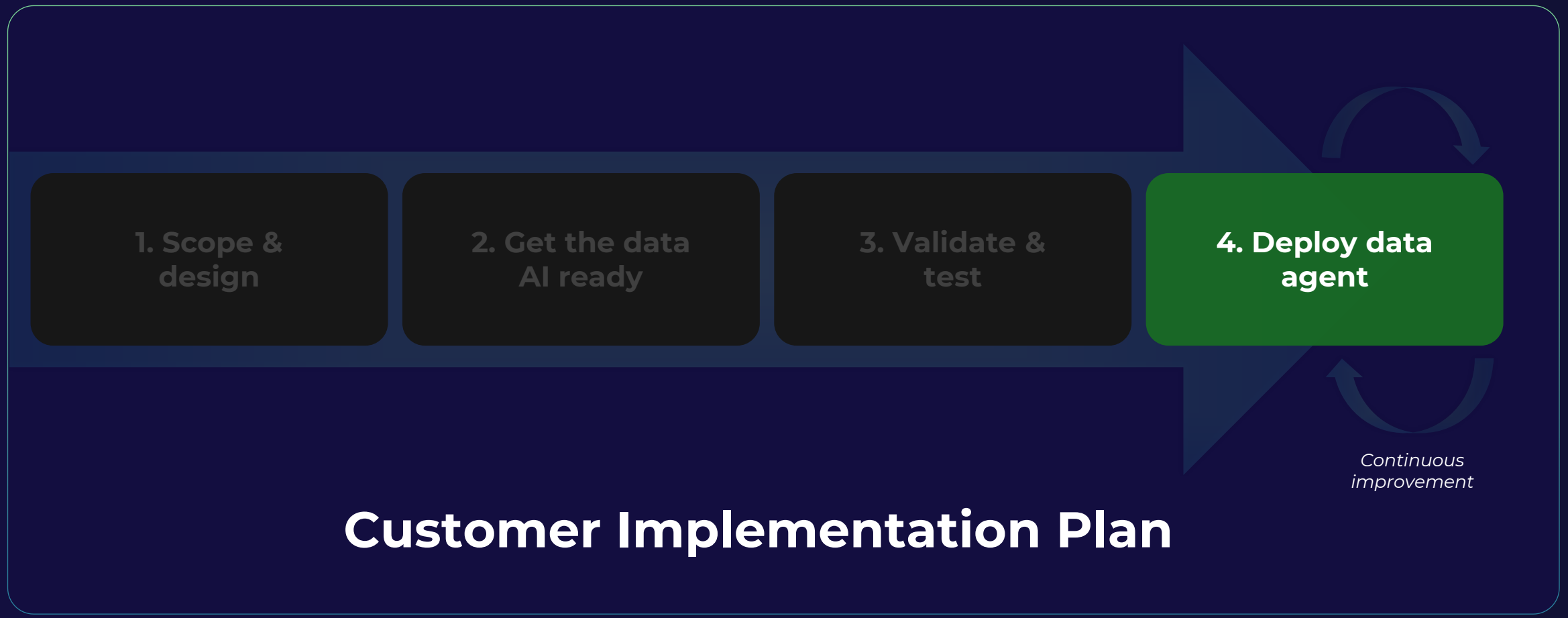
- Query the agent with typical, edge, and out-of-scope queries.
- Debugging: Review agent behavior in depth (steps, generated SQL/DAX/KQL).
- Evaluate agent on factual accuracy, relevance, helpfulness, compliance.

Optimization actions:

- Refined instructions or example queries.
- Adjusted data source prioritization.
- Pre-aggregate or index data for performance.

Implement CI/CD strategy

Planning a Successful Data Agent Implementation



Customer Implementation Plan

Agent readiness checklist

Implement source control

- All agent artifacts (instructions, data source configs, example queries) are version-controlled.
- Draft vs. Published version strategy is defined.
- Change log is maintained.
- CI/CD process includes schema validation, regression testing, and promotion steps.

Fine-tune Instructions

- Agent instructions include objective, data source priority, terminology, formatting, and handling rules. Instructions are tested for clarity and effectiveness.
- Only relevant tables exposed. Table/column names are business-friendly.
- Data source instructions include table descriptions, joins, and query logic.
- Example queries added for each supported source.
- Power BI semantic models include validated measures, hierarchies, and descriptions.

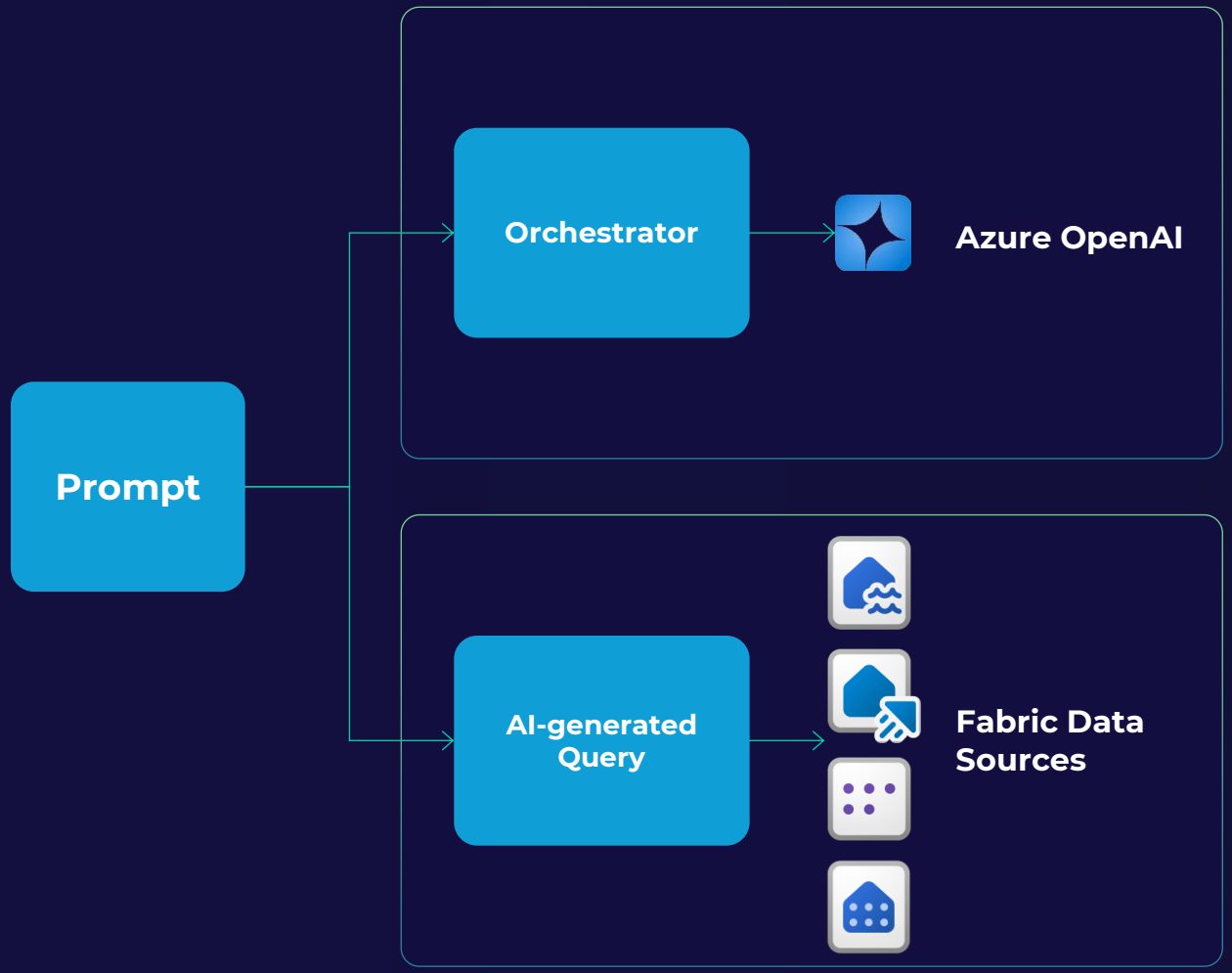
Secure the agent's data

- RBAC and RLS configured and tested.
- Sensitive data masked or excluded.
- Agent queries run under user identity with correct permissions.

Create adoption & lifecycle strategy

- Agent accessible in intended channels (Fabric, Power BI, Teams).
- Quick-start guide or training provided.
- Feedback loop established.
- Agent owner assigned.
- Maintenance plan in place (updates, feedback, schema monitoring).
- Retirement criteria defined.

Understanding Fabric data agent billing

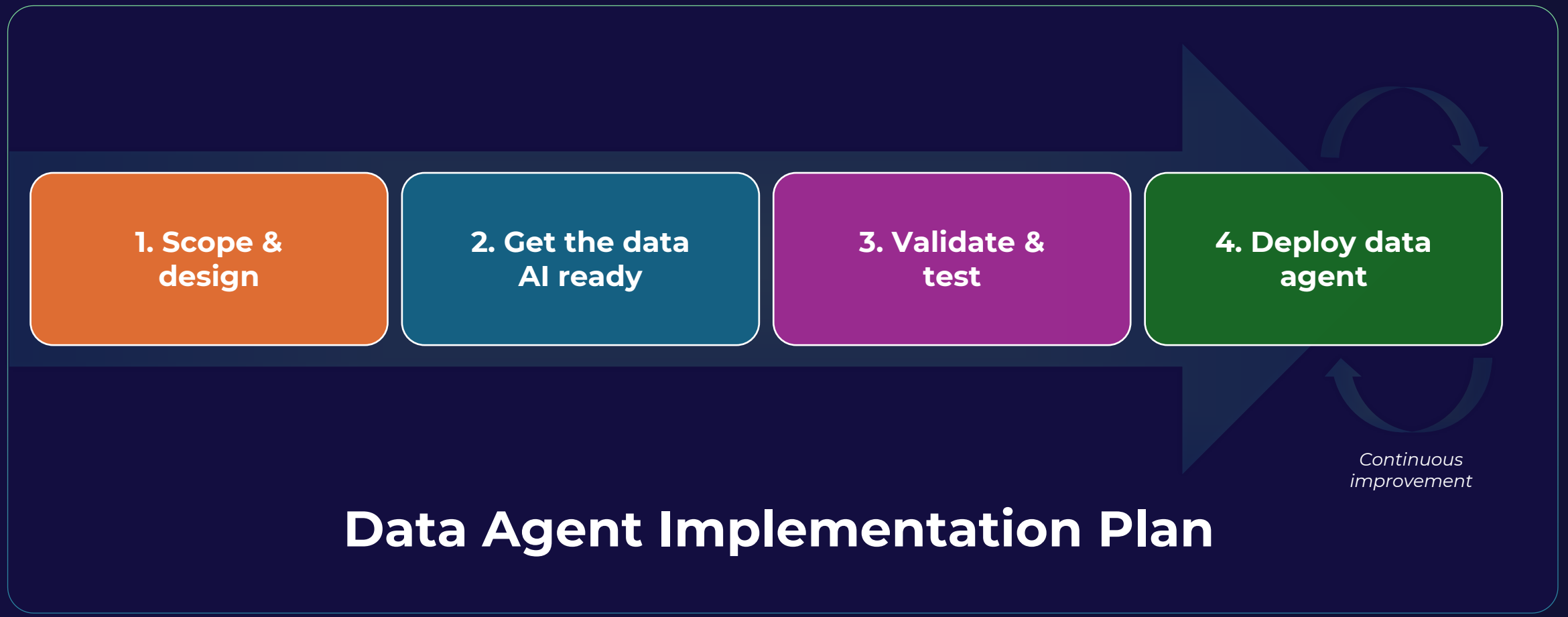


Leverages Azure OpenAI models to process and generate responses

AI Query is measured by number of tokens processed

DAX & KQL queries generated for data source to answer questions are billed separately (SQL queries are billed within Data Agent CU)

Planning a Successful Data Agent Implementation



Q&A

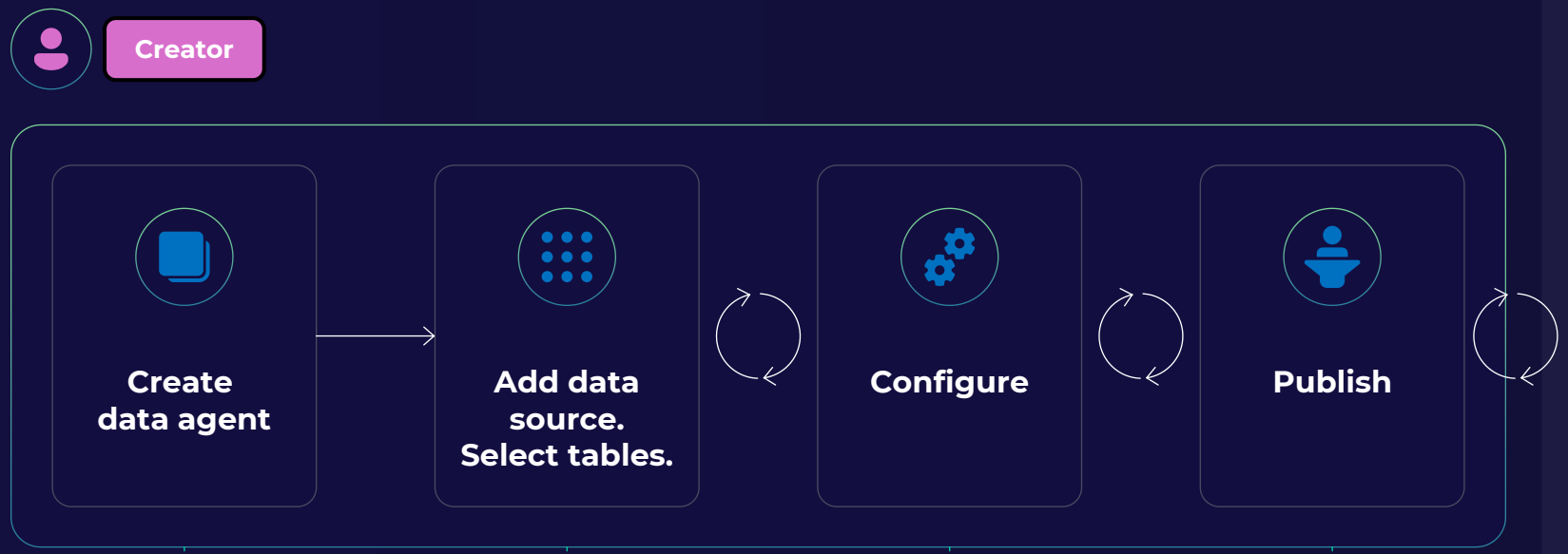
Please submit questions at: https://aka.ms/dataagent-qa



Consumer Experiences in Data Agent

Data Agent end-to-end flow

End-to-end scenario



Create new item type in Fabric:

Track lineage

Manage permissions

Share

Select data source:

Lakehouse

Warehouse

Semantic model

Eventhouse KQL DB

Mirrored Databases

Configure:

Instructions for AI

Few shot examples

Publish:

Creates a published version

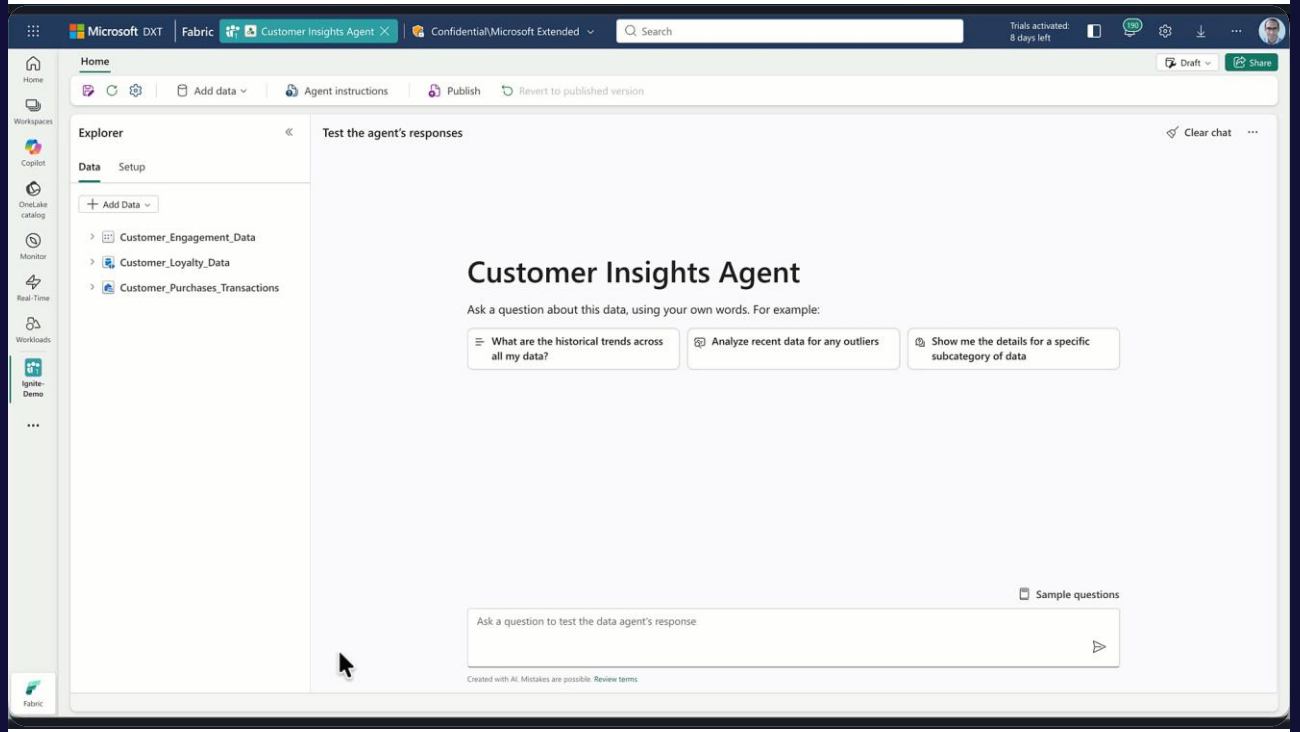
Can be shared with consumers

Fabric data agent integration with M365

Fabric data agents can reason over and synthesize data in OneLake for insights

AI creators can enrich M365 Copilot with Fabric data agents' data domain expertise

Security permissions, including RLS and CLS, are always respected



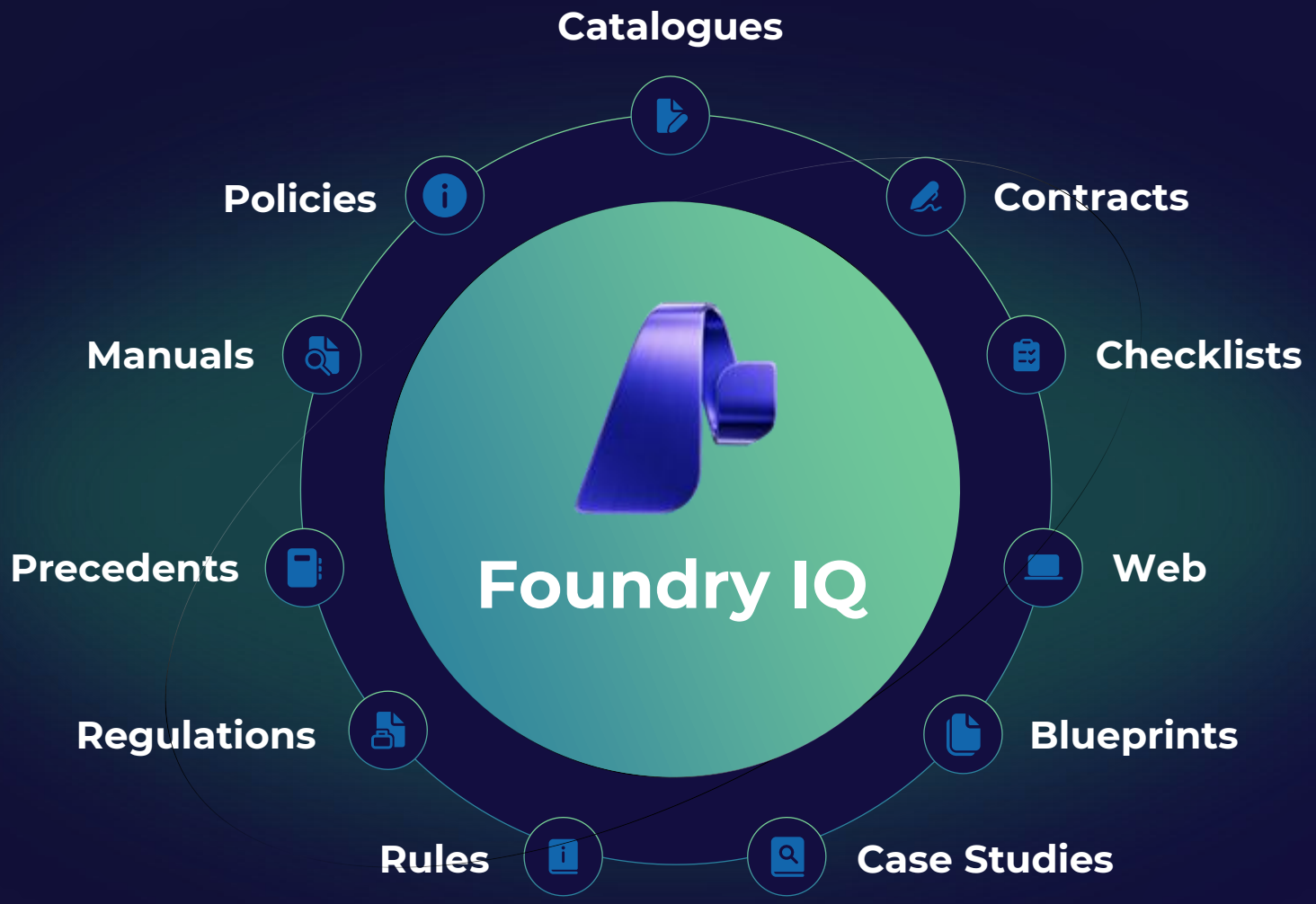
M365 – Data Agent Demo

Q&A

Please submit questions at: aka.ms/dataagent-tutorial-qa

Please provide feedback on the tutorial:
aka.ms/dataagent-tutorial-feedback

Unstructured Data in Data Agent



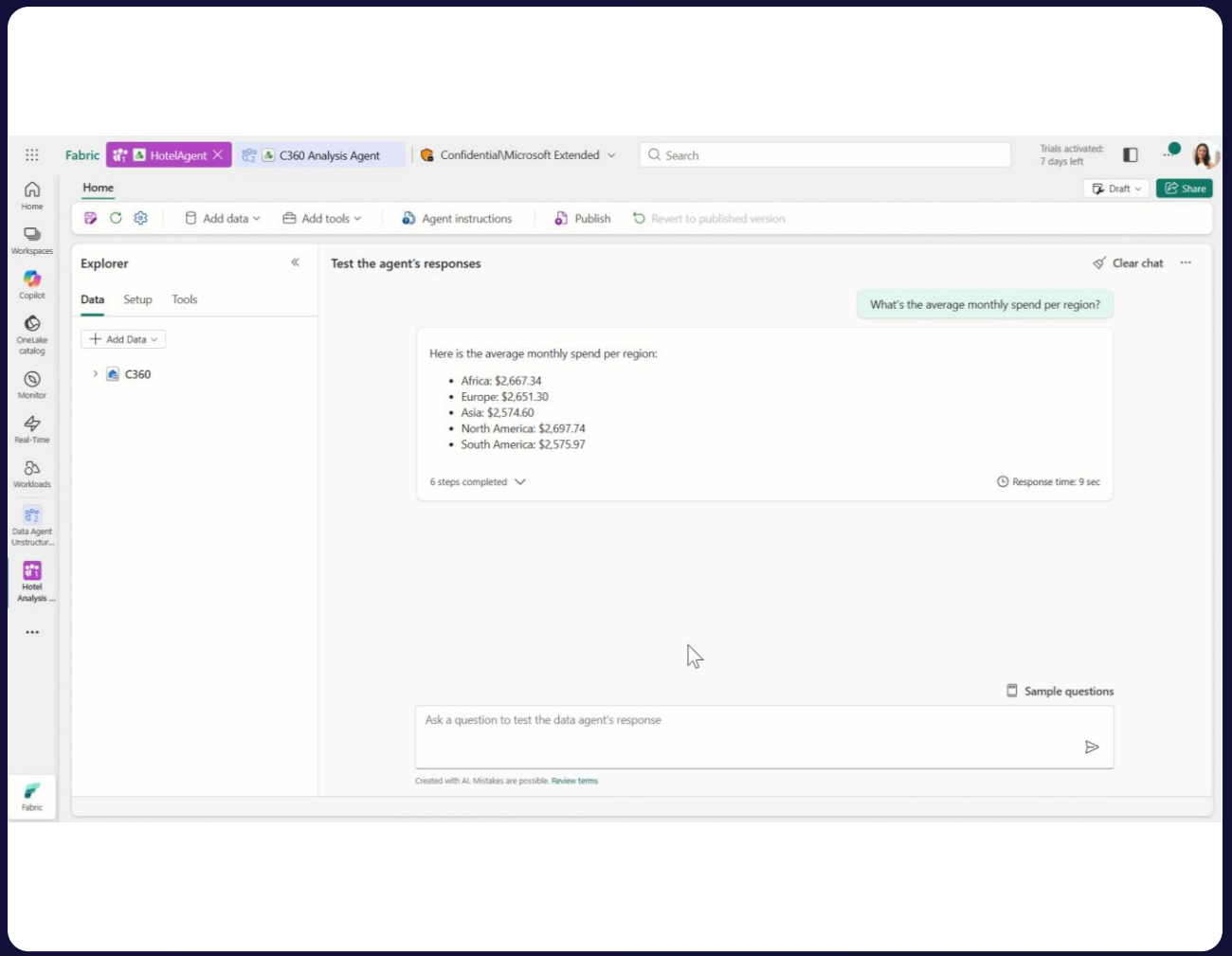
Microsoft Foundry: Context on policies, procedures, and institutional knowledge

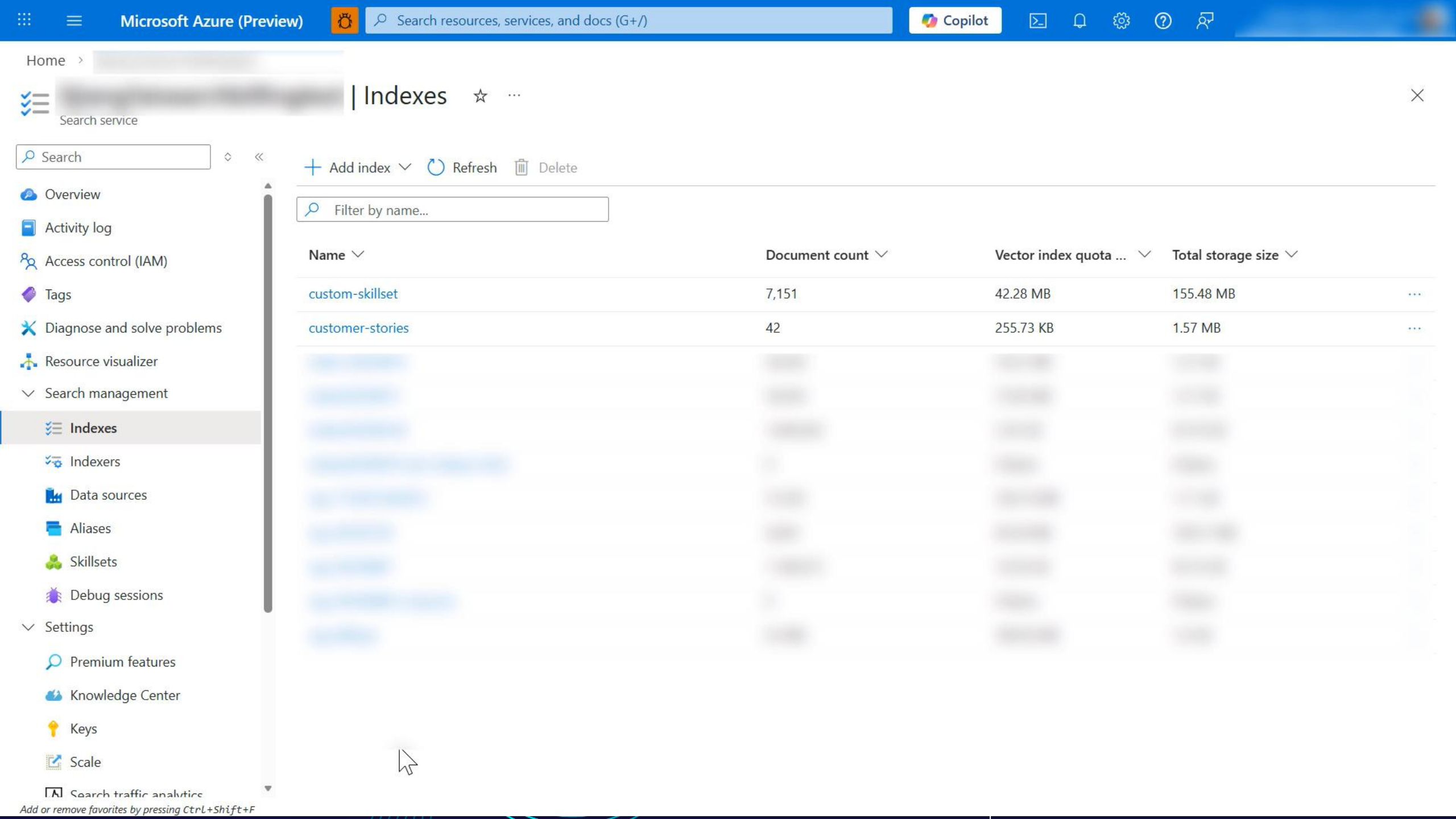
Bring your own Foundry AI Search Index

Bring your own Foundry AI Search Index to a Fabric Data Agent

Use the built-in AI Search tool for custom indexes

Ask natural language questions over your AI Search index and join with other data sources




Microsoft Azure (Preview)
Search resources, services, and docs (G+/)
Copilot
Home
| Indexes
Search service
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Search management
Indexes
Indexers
Data sources
Aliases
Skillsets
Debug sessions
Settings
Premium features
Knowledge Center
Keys
Scale
Add index
Refresh
Delete
Filter by name...
Name
Document count
Vector index quota ...
Total storage size
custom-skillset
7,151
42.28 MB
155.48 MB
customer-stories
42
255.73 KB
1.57 MB
Add or remove favorites by pressing Ctrl+Shift+F

Lab 4: Unstructured Data

Auditing Data Agent Interactions

Viewing Prompts

"As an admin, Can I see what my users are asking Data Agent?"

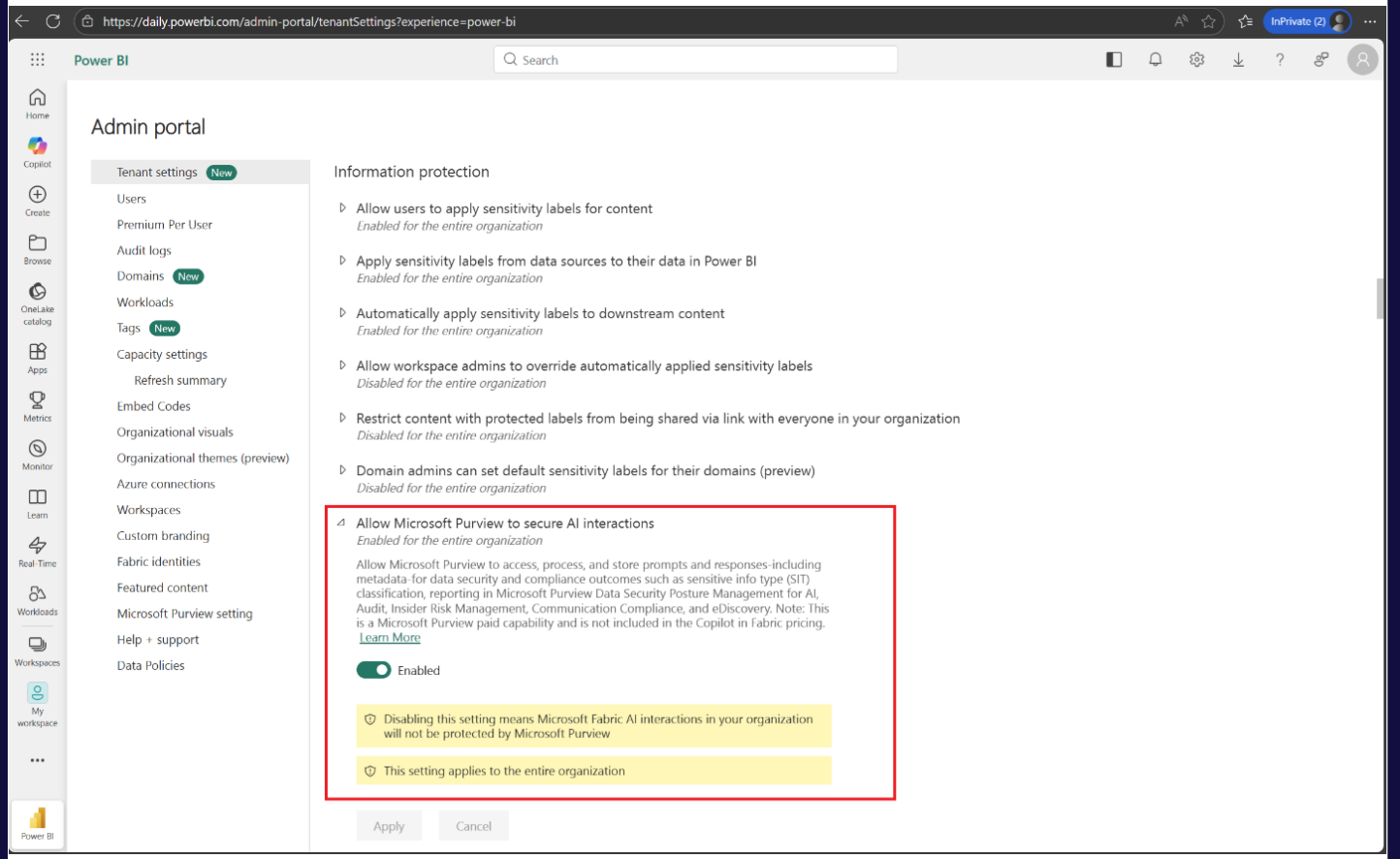
Fabric does not natively offer an option to review a user's prompts or responses from Data Agent

Purview DPSM (Data Posture Security Management) for AI does!

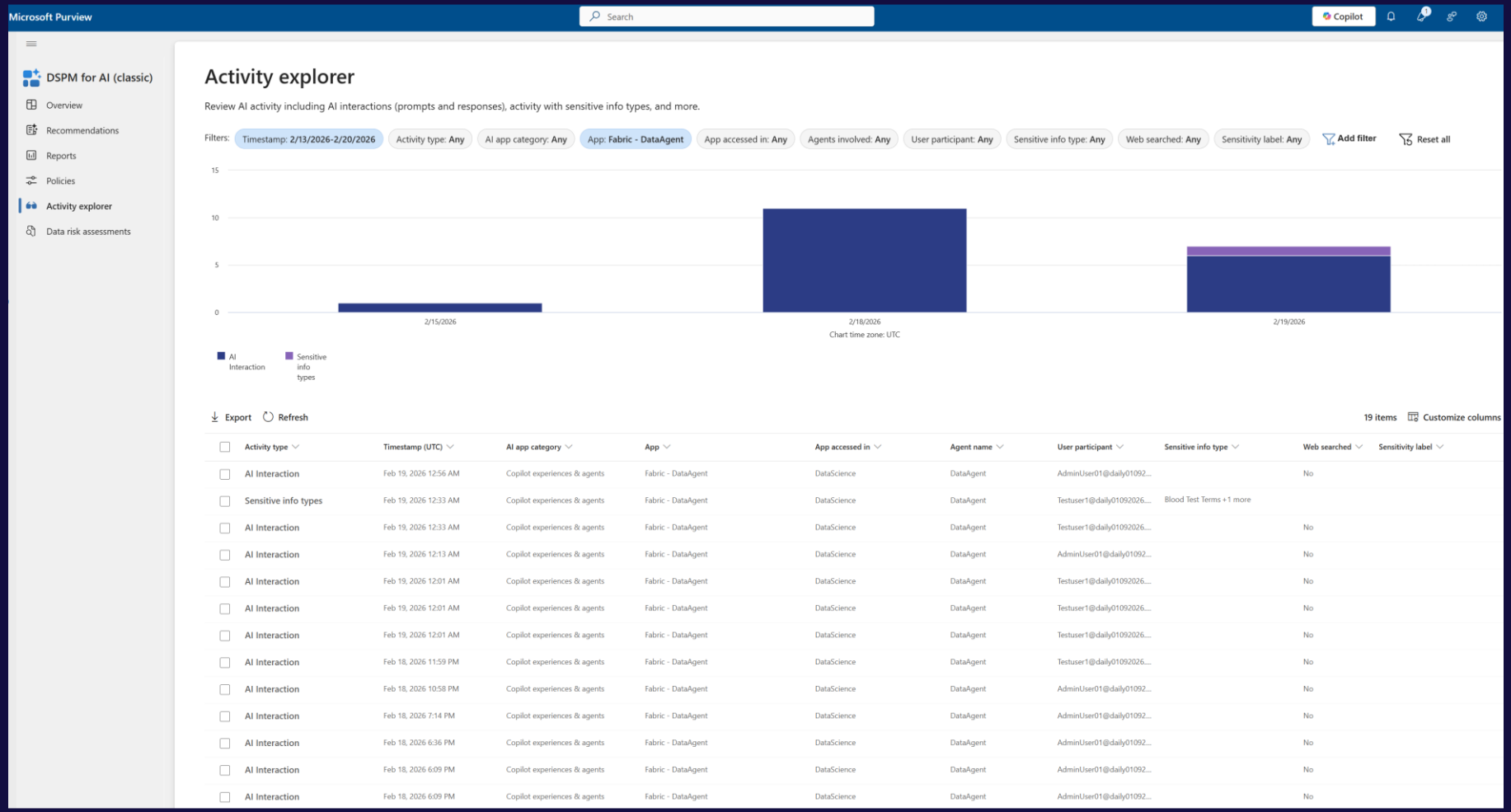
- Security and Governance for the AI era
- Requires separate licensing for Purview! (docs)
- Collection Policy "DSPM for AI-Capture interactions for Copilot experiences"

Tenant admins with Purview licenses can allow Microsoft Purview to secure AI interactions by enabling this setting within Fabric

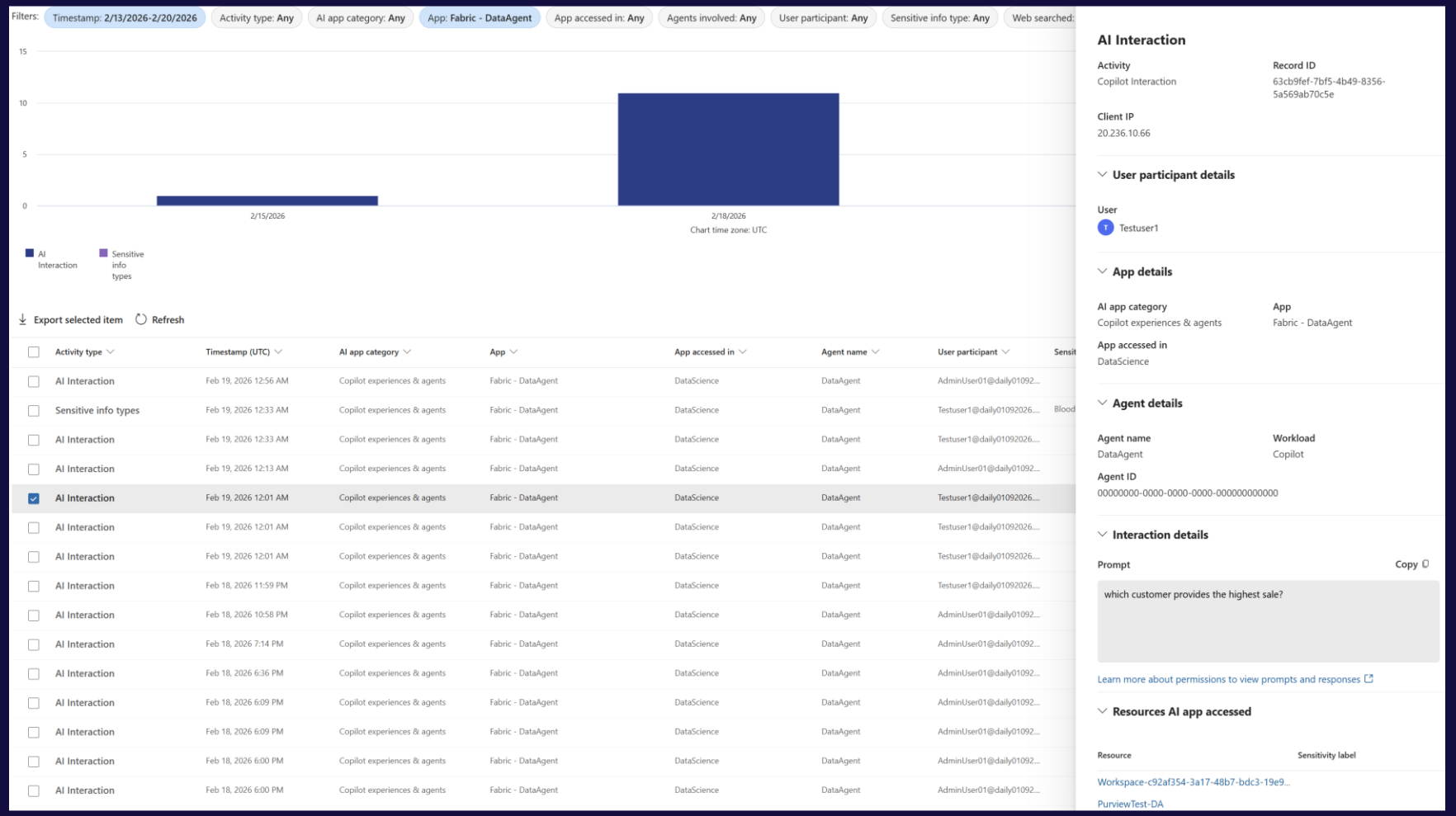
When a user performs an action within Fabric Copilots or Data Agent, an audit log entry will be created



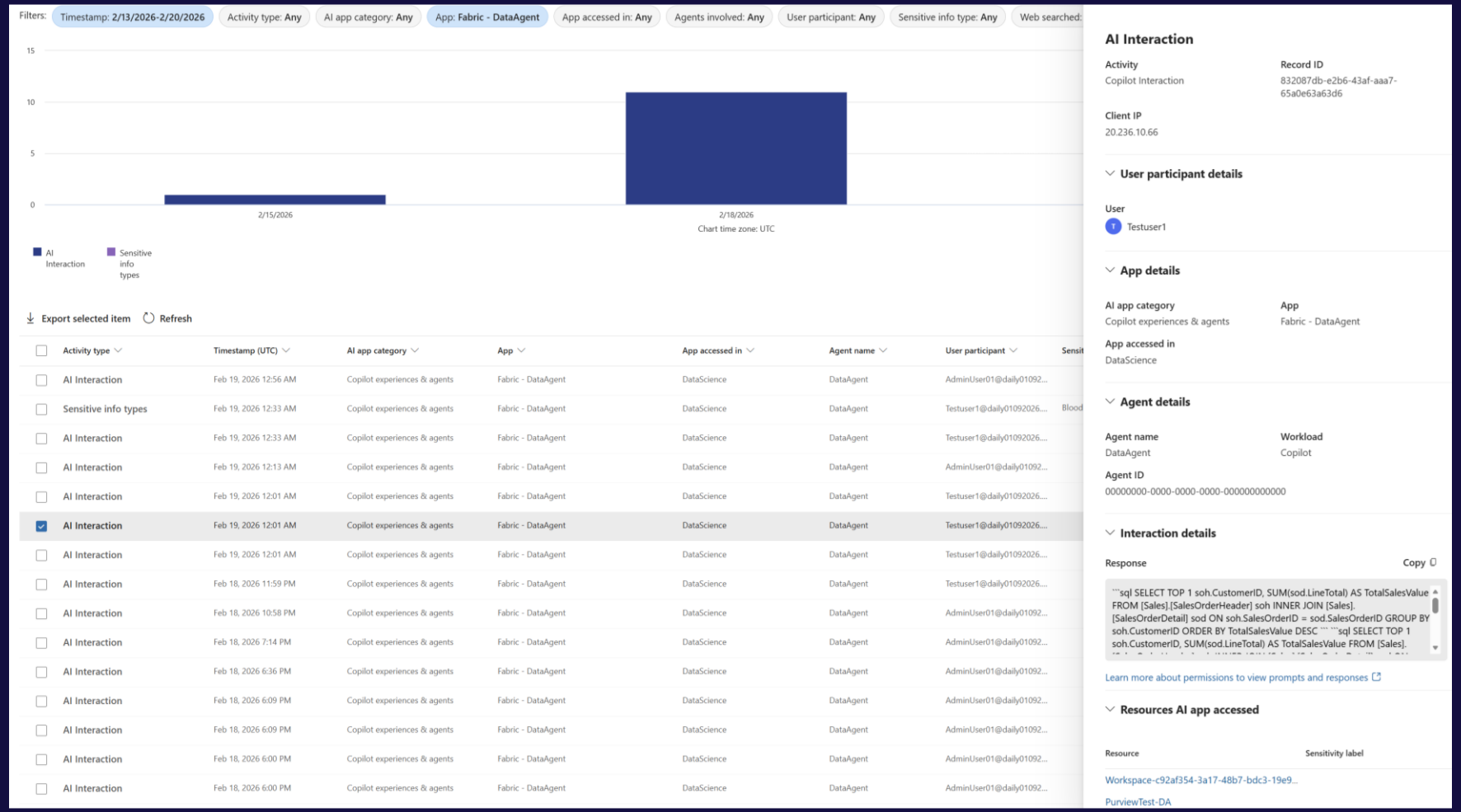
Activity Explorer – DSPM for AI



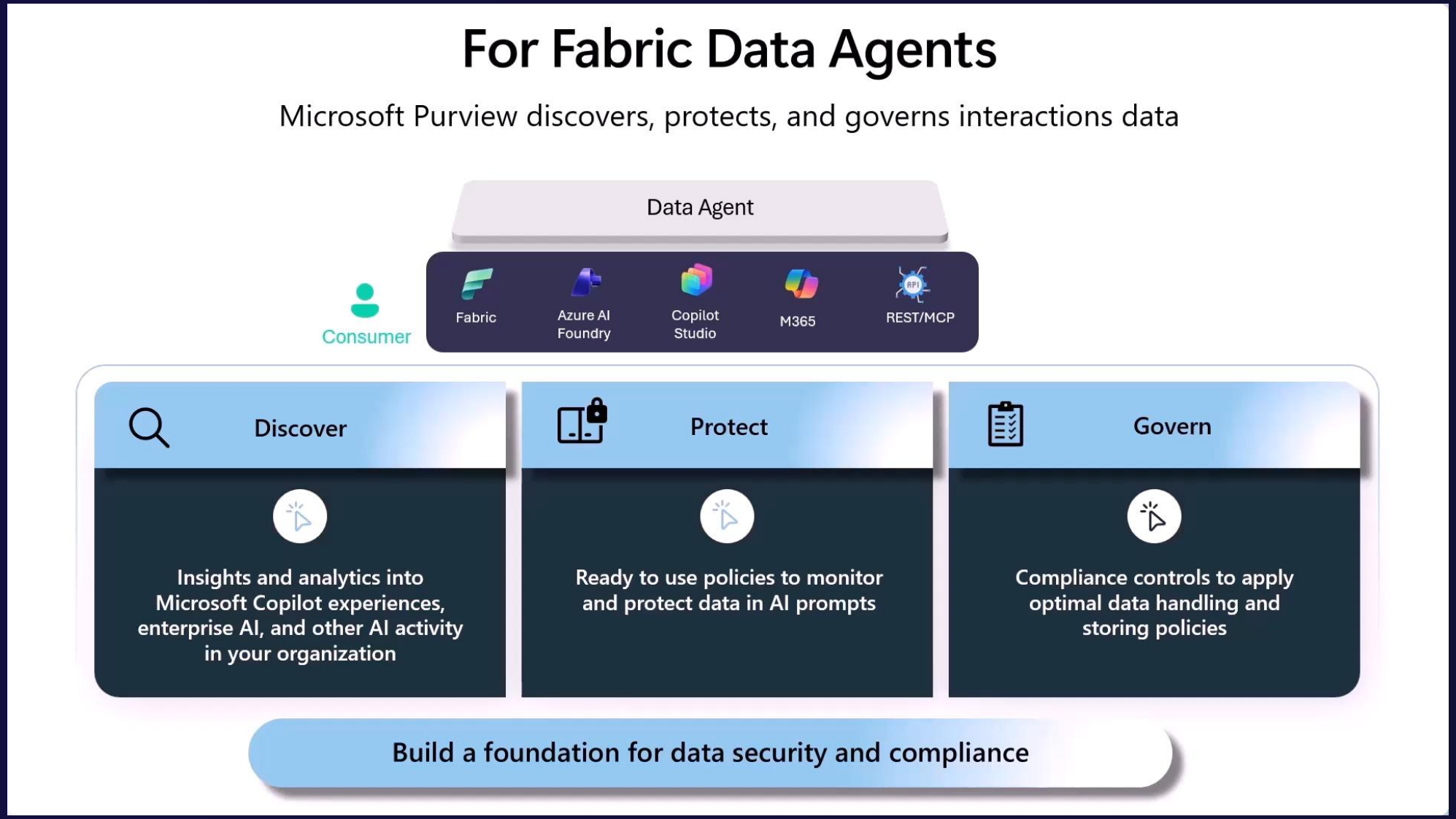
Data Agent Prompt



Data Agent Response



Demo


For Fabric Data Agents
Microsoft Purview discovers, protects, and governs interactions data
Data Agent
Consumer
Fabric
Azure AI Foundry
Copilot Studio
M365
REST/MCP
Discover
Insights and analytics into Microsoft Copilot experiences, enterprise AI, and other AI activity in your organization
Protect
Ready to use policies to monitor and protect data in AI prompts
Govern
Compliance controls to apply optimal data handling and storing policies
Build a foundation for data security and compliance

Don’t Miss our other session throughout the conference!

Date	Time	Session Title	Speaker(s)
Wednesday, Mar 18, 2026	1:45 PM – 2:45 PM	CORENOTE: The Future of AI in Microsoft Fabric: Data Agents and Beyond	Nellie Gustafsson; Markus Weimer
Wednesday, Mar 18, 2026	3:05 PM – 4:05 PM	Unlocking Copilot in Fabric: Administration, Governance, and Beyond!	Dan English; Sandeep Pawar
Thursday, Mar 19, 2026	8:00 AM – 9:00 AM	AI Answers, Done Right: When to use Power BI Copilot vs. Fabric Data Agents	Sara Vredevoogd; Carly Newsome
Thursday, Mar 19, 2026	10:10 AM – 11:10 AM	Building and Deploying Smarter Data Agents in Microsoft Fabric	Misha Desai; Amir Jafari
Thursday, Mar 19, 2026	2:00 PM – 3:00 PM	Expert vs. AI: The Power BI Turing Test	Sujata Narayana; Sid Vemuri
Thursday, Mar 19, 2026	2:00 PM – 3:00 PM	Powering Data with AI Functions in Fabric	Virginia Roman; Rana Singh
Friday, Mar 20, 2026	11:30 AM – 12:30 PM	Reporting with action: Visualizations with translytical task flows in Power BI and Fabric	Zoe Douglas; Denyse Niwenshuti
Friday, Mar 20, 2026	2:00 PM – 3:00 PM	Beyond Monitoring: AI-Driven Spark Optimization in Microsoft Fabric	Anu Venkataraman; Virginia Roman
Friday, Mar 20, 2026	2:00 PM – 3:00 PM	Forget Prompts: Building Declarative AI Pipelines in Microsoft Fabric	Sandeep Pawar

Q&A

Please submit questions at: https://aka.ms/dataagent-qa

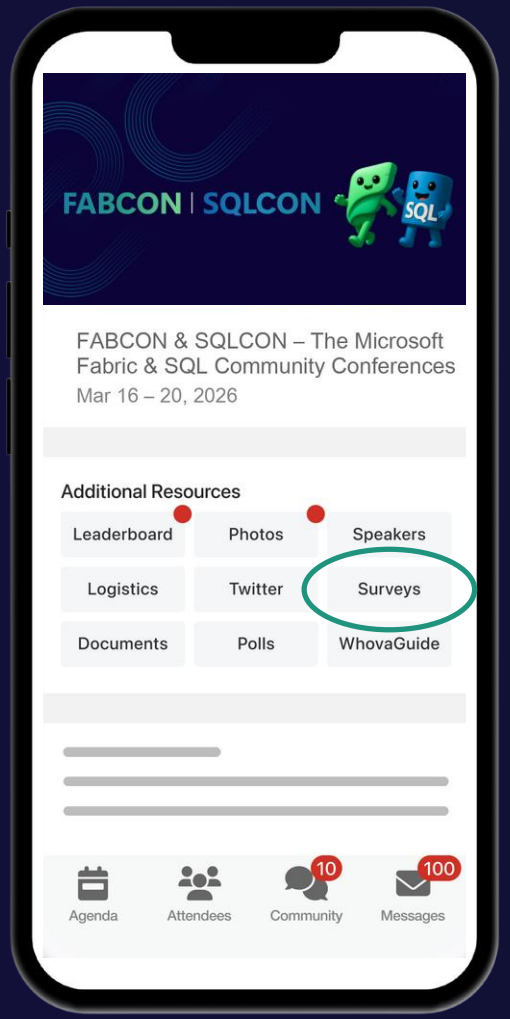


Please Provide feedback for this session:
https://aka.ms/dataagent-tutorial-feedback



How was the session?

Complete Session Surveys in Whova for your chance to WIN PRIZES!



End of Tutorial